Effects of Subsidies on Welfare and Market Structure in the U.S. Broadband Industry*

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Abstract

How should governments split broadband subsidies between consumer discounts and infrastructure grants? I study the Affordable Connectivity Program (ACP) and the Broadband Equity, Access, and Deployment (BEAD) program using state-year event studies and a tract-level structural model with endogenous entry, estimated on 220,821 provider-tier products in 69,085 census tracts from FCC, URS, and ACS data. Event studies show that ACP was followed by lower prices, faster plan mixes, and lower concentration in high-eligibility states; BEAD effects are smaller and delayed, consistent with construction lags. Counterfactuals imply a BEAD benefit-cost ratio of 4.0–12.3 and an ACP ratio of 0.47–1.03 under targeted income eligibility. ACP delivers the greatest returns where broadband exists but remains unaffordable; BEAD delivers them where infrastructure is missing. The programs are complements, not substitutes.

JEL classification: D22, L13, L51, L52, L96, L98.

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1 Introduction

Broadband access has become a prerequisite for economic participation, yet the digital divide remains wide. In 2020, roughly 21 million American households lacked fixed broadband service, a gap that falls disproportionately on low-income and rural communities.[1] Congress responded with the Infrastructure Investment and Jobs Act (IIJA) of 2021, which allocated approximately \$65 billion to broadband and created two flagship programs: the Affordable Connectivity Program (ACP), a \$14.2 billion demand-side subsidy providing eligible low-income households with monthly discounts of up to \$30 on broadband bills, and the Broadband Equity, Access, and Deployment (BEAD) program, a \$42.45 billion supply-side grant funding deployment in unserved and underserved areas. The two programs share a goal—wider, more affordable broadband—but attack different constraints. ACP lowers the consumer-facing price where service already exists. BEAD lowers the fixed cost of building service where private deployment has not materialized.

In this paper, I ask three questions. First, did ACP and BEAD change broadband prices, product offerings, and local market structure after IIJA? Second, how large are the welfare gains from each program once firms can enter or adjust local product positions? Third, are demand-side and supply-side subsidies substitutes, or do they address different parts of the digital divide? These questions matter for current policy. ACP exhausted its funding in June 2024, leaving policymakers to decide whether a successor affordability program is worth financing. BEAD is the largest federal broadband deployment program in U.S. history, but its returns depend on whether subsidies induce entry in markets where deployment is privately unprofitable. Standard evaluations that count new subscribers or dollars per connection do not answer these questions, especially in local broadband markets with few providers and substantial infra-marginal households.

To answer them, I combine reduced-form and structural evidence. The reduced-form analysis uses continuous-treatment event studies to compare states with different exposure to ACP eligibility and BEAD funding before and after IIJA—a design well suited to documenting the timing and direction of observed policy effects. The structural model then translates those effects into welfare and counterfactual policy designs. I estimate a nested-logit demand system, recover Bertrand-Nash markups and marginal costs, and bound fixed costs of entry using the scalable discrete-game method of Fan and Yang [2024]. This framework fits the policy setting naturally: broadband subsidies change both the demand margin and product availability, so a model that treats entry and pricing jointly is the right tool. The discrete-game approach

[1] Federal Communications Commission [2020] reported 21.3 million Americans lacking access to fixed broadband at 25/3 Mbps speeds based on FCC Form 477 data (see page 19).

also avoids having to solve the entry game inside the estimation loop, which matters when there are tens of thousands of markets and many potential provider-tier positions.

The data combine several official federal sources. For the reduced-form analysis, I build a balanced panel of 52 U.S. states from 2014 to 2025 using FCC Urban Rate Survey prices and product characteristics, ACP eligibility exposure from the American Community Survey, and BEAD allocations from federal program records. For the structural analysis, I construct a pre-IIJA product-level dataset for December 2020. The demand sample contains 220,821 observed broadband products in 69,085 census tracts, defined at the tract-provider-speed-tier level. The entry sample augments these observed products with feasible but unobserved county-footprint positions. Product characteristics come from FCC Form 477 deployment records, prices from the FCC Urban Rate Survey, adoption and outside-option shares from FCC broadband adoption data and the American Community Survey, and demographics from American Community Survey tract files. No existing dataset assembles comparable coverage of broadband market conditions—availability, prices, technology, demographics, adoption, and potential entry positions—at this geographic and product detail, pre-dating both IIJA programs.

The results show that the programs operate through different margins. In the event study, ACP is associated with lower prices, faster plan mixes, and lower concentration in high-eligibility states. Moving from the 25th to the 75th percentile of ACP eligibility exposure is associated with average broadband prices roughly 7–10 percent lower at peak response, with effects on high-speed plans reaching 15–18 percent. BEAD effects are smaller and appear with a two-to-three year lag, consistent with the time required for infrastructure deployment. In the structural counterfactuals, a 75 percent BEAD fixed-cost subsidy induces 116,015–339,155 new provider-tract entries and raises consumer surplus from \$1.80–\$2.19 billion per month to \$2.36–\$2.77 billion. A \$30 ACP discount raises consumer surplus to \$2.59–\$2.71 billion per month for eligible households, with a cost-benefit ratio of 0.47–1.03 under targeted eligibility. BEAD generates a BCR of 4.0–12.3. The central lesson is that the programs are complements: ACP is most useful where infrastructure exists but affordability limits adoption; BEAD is most useful where infrastructure is missing.

The paper speaks to three related literatures. On broadband markets and the digital divide, existing work shows that adoption depends on prices, income, local market structure, and provider choice [Rosston et al., 2010, Nevo et al., 2016, Flamm and Varas, 2022, Espín and Rojas, 2024, Kearns, 2024]; I build on that evidence by using policy exposure as the source of identifying variation and by tracing market-structure responses that aggregate adoption studies cannot capture.[2] On structural methods for differentiated-product markets with endogenous entry, the approach is closest to Tamer [2003], Ciliberto and Tamer [2009], Ciliberto et al. [2021], and Fan and Yang [2024], and is the first application of scalable discrete-game bounds to broadband infrastructure. On the evaluation of public subsidies in imperfectly competitive markets, the paper is unusual in placing a demand-side transfer and a fixed-cost deployment subsidy inside the same welfare accounting exercise, which allows a direct fiscal-efficiency comparison. The results speak to a growing literature on infrastructure crowd-out [Wilson, 2025]; because I do not model that channel, the BEAD welfare estimates reflect the return to subsidized private entry, not to public provision.

The rest of the paper proceeds as follows. Section 2 describes the institutional setting, data, and local market structure. Section 3 presents the reduced-form evidence. Section 4 develops the structural model, including timing assumptions, identification, and a discussion of multiple equilibria. Section 5 reports estimation results and structural parameter diagnostics. Section 6 presents counterfactual policy simulations and distributional effects across market types. Section 7 reports the cost-benefit analysis, including sensitivity to discount rate and infrastructure horizon. Section 8 concludes.

2 Industry Background and Data

2.1 Industry Background

2.1.1 Current Landscape

Residential broadband is technologically fragmented in a way that shapes both competition and the geography of access. Service reaches households over four distinct platforms—DSL on legacy copper, cable modem on coaxial television networks, fiber optics, and fixed wireless—and these platforms differ sharply in speed, reliability, and the upfront capital they require. Fiber delivers the highest performance but demands large sunk investment; fixed wireless economizes on construction in rural areas where wired deployment is uneconomic. Providers compete on speed, data allowances, and price. Telephone and cable incumbents still dominate, but national fiber entrants, fixed-wireless operators, and a handful of municipal systems now contest local markets.[3]

[2] Concentration, high prices, and limited provider choice are well documented in U.S. broadband [Flamm and Varas, 2022]. Downstream consequences for labor markets, education, and health are documented in Forman et al. [2005], Goldfarb and Prince [2008], and Goolsbee and Klenow [2002].

[3] Wilson [2025] finds that municipal entry crowds out private ISP investment, an important caveat for evaluating BEAD's net effect on private deployment. See Appendix A for an overview of broadband technology types.

Two facts define the market and explain why Congress intervened: service is expensive relative to household income, and most households cannot choose among more than two or three providers [Flamm and Varas, 2022]. Both hold sharply in my December 2020 data. The average census tract is served by just 2.36 fixed-broadband providers, and the median within-tract provider HHI is 5,712—more than double the 2,500 threshold conventionally associated with concentrated markets. The shortfall is not spread evenly: fiber, the platform with the best performance, concentrates in cities and suburbs, while rural households still depend on aging copper DSL whose speeds and reliability fall well short of the federal broadband standard.

These conditions produce the digital divide: a gap in reliable internet access that tracks income and geography closely and whose consequences run through job search, healthcare, and schooling [Forman et al., 2005, Goldfarb and Prince, 2008, Goolsbee and Klenow, 2002]. Federal policy has responded along the same two margins that define the underlying failures—lowering the price of service that already exists, and paying to build service where private providers have not.

2.1.2 Federal Broadband Subsidies

The 2021 Infrastructure Investment and Jobs Act (IIJA) committed roughly \$65 billion to broadband through two programs that attack opposite sides of the market. The Affordable Connectivity Program (ACP), funded at \$14.2 billion, subsidized household bills directly. The Broadband Equity, Access, and Deployment (BEAD) program, funded at \$42.45 billion, subsidizes deployment in underserved areas.

ACP succeeded the Emergency Broadband Benefit (EBB), which reached nearly nine million low-income households at up to \$50 per month between February and December 2021. ACP scaled far larger, enrolling 23 million households at its peak with a monthly discount of up to \$30 (\$75 on qualifying Tribal lands) and a one-time device discount of up to \$100.[4] Its funding lapsed in June 2024 when Congress declined to reappropriate. This full lifecycle—launch in 2022 through exhaustion in 2024—falls entirely inside the reduced-form event-study window, and its abrupt end sharpens the central policy question: is a successor affordability program worth financing, and in what form? The welfare and fiscal-efficiency analysis below speaks directly to that question.

BEAD is the largest federal program ever to subsidize broadband providers directly. Funds are allocated to states by formula on the count of underserved locations—broadband-serviceable locations lacking service entirely or lacking reliable service at 25/3 Mbps with

[4] https://www.fcc.gov/acp

latency below 100 milliseconds.[5] Because the formula is mechanical, no state can influence its own allocation, which makes the variable a clean predetermined treatment. The program's slow institutional clock explains the lagged effects documented below: IIJA passed in November 2021, but allocations were not announced until June 2023, initial state proposals were due in December 2023, and subgrantee selection, permitting, and construction follow only afterward. Deployment is expected to extend through 2026 and beyond. The structural counterfactuals, anchored to a pre-IIJA baseline, are therefore best read as the program's steady-state effects once deployment is complete, not as forecasts for any single intermediate year.

2.1.3 Economic Rationale

Each program corrects a distinct market failure, and the distinction is the organizing idea of the paper. On the demand side, affordability binds: even where service exists, low income and limited digital literacy keep many households below the participation margin. In 2020, roughly 15 percent of U.S. households with broadband access reported being unable to afford their plan, and subscription rates among households below 200 percent of the federal poverty line ran about 20 percentage points below the national average. A demand-side subsidy attacks this gap directly, lowering the effective price until price-constrained households subscribe.

On the supply side, deployment binds: fixed costs are high enough to make broadband privately unprofitable in low-density or low-income areas even when the social return to connectivity is large. This wedge between private and social returns to infrastructure is the classic case for public subsidy [Bresnahan and Reiss, 1991, Kearns, 2024]. The case is not unqualified—Wilson [2025] shows that public supply-side intervention can crowd out private investment, a general-equilibrium cost most evaluations ignore. BEAD lowers the provider's private cost of entry, but its net effect depends on how much private deployment it displaces.

The two failures are largely non-overlapping, which makes the programs complements rather than substitutes. A demand subsidy does nothing where no infrastructure exists: in wholly unserved areas, no discount can induce adoption of a product that is not offered. A deployment subsidy has limited reach on affordability where competitive infrastructure already exists and price, not access, is the binding constraint. In monopoly or duopoly markets that are technically served but where a second provider would face high entry costs, BEAD-induced entry can discipline prices through Bertrand competition—so the distinction is sharpest in markets that are fully served and competitive, and less sharp at the coverage margin. Each program therefore addresses the failure the other cannot in most markets, with some overlap at the frontier.

[5] https://broadbandusa.ntia.doc.gov

Market structure adds a further channel. With most households facing one to three providers [Flamm and Varas, 2022], broadband is an oligopoly, and a demand subsidy that enlarges the addressable market can move posted prices and product menus—equilibrium responses that subscriber counts miss. The reduced-form event study tests for these responses; the structural counterfactuals then hold gross provider prices fixed, isolating the demand-expansion channel from provider price adjustment.

2.2 Data Sources and Estimation Samples

The empirical analysis draws on five publicly available data sources. Each is described below with its collection period, information content, and unit of observation.

Federal Communications Commission Urban Rate Survey. The FCC collects the Urban Rate Survey (URS) annually from a stratified random sample of ISPs operating in urban census tracts, spanning 2014 to 2025.[6] Each record reports the plan's advertised download and upload speeds, technology type, monthly price, and a plan-availability weight. URS prices and speed characteristics are the primary source of product-level broadband prices in both the state-year panel and the structural estimation sample.

FCC Form 477 Deployment Data. Form 477, collected semi-annually, records where ISPs offer fixed broadband service, spanning 2014 to 2023 at the provider-census-block level.[7] I restrict to residential fixed broadband and use the December 2020 release as the structural baseline. Form 477 records define product—which provider offers which speed tier in which census tract—and are the basis for constructing observed and potential products entering the entry game.

American Community Survey. The American Community Survey provides annual demographic and socioeconomic estimates at the census-tract level.[8] From the 2020 five-year release I extract housing units (market size), median household income, poverty rate, broadband subscription rate, renter share, college-educated share, and the age-65+ share; from the one-year release, the share of workers employed from home. These variables serve as market-size measures, eligibility proxies, and demand-side controls.

[6] Data available at: `https://www.fcc.gov/economics-analytics/industry-analysis-division/urban-rate-survey-data-resources`.
[7] Data available at: `https://broadband477map.fcc.gov/#/data-download`.
[8] Data available at: `https://www.census.gov/programs-surveys/acs`.

FCC Broadband Adoption Data. The FCC publishes tract-level fixed broadband adoption statistics derived from Form 477 subscription filings, reporting the share of households with fixed broadband connections by speed threshold in binned ranges.[9] I use the December 2020 file to anchor the product-level market share construction described in Section 2.5.[10]

BEAD Program Allocation Records. State-level BEAD allocations, announced in June 2023, are published by the NTIA and determined by a federal formula based on certified counts of unserved broadband-serviceable locations.[11] Because states cannot influence their allocation amount, this variable serves as a predetermined continuous treatment in the event study.[12]

These five sources feed two estimation samples, each matched to a different question. The *state-year policy panel* covers 52 states (including the District of Columbia and Puerto Rico) annually from 2014 to 2025 and supports the event study, pairing Urban Rate Survey price and plan-composition outcomes with BEAD allocation intensity and American Community Survey poverty rates. The *tract-level structural sample* is a single December 2020 cross-section—220,821 observed provider-tier products in 69,085 tracts—and supports demand estimation, marginal-cost recovery, and fixed-cost bounds, combining Form 477 availability, Urban Rate Survey prices, FCC adoption shares, and American Community Survey demographics.

Data quality, not only identification, dictates this split. A consistent tract-year panel cannot be built cleanly over 2014–2025 because the FCC's post-2021 transition from Form 477 to the Broadband Data Collection system redefined geographic units and coverage measurement, which would inject spurious dynamics into any long tract panel. The state-year aggregation absorbs that break, and the single-cross-section structural model sidesteps it entirely. Appendix B reports file-level details, merge keys, and sample restrictions; the full replication package is available upon request.

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[9] The December 2020 release is available at: `https://docs.fcc.gov/public/attachments/DOC-395959A1.pdf`.

[10] The 2020 FCC adoption file reports tract-level subscription shares only at the 10/1 Mbps threshold, not at 25/3 Mbps. The 10/1 Mbps bin is nonetheless a close proxy for 25/3 Mbps adoption. Provider plan menus in 2020 clustered below 10 Mbps (legacy DSL) or above 25 Mbps (cable and fiber), with very few plans between the two thresholds [Federal Communications Commission, 2022], so subscribers at $\geq$10/1 Mbps and subscribers at $\geq$25/3 Mbps are nearly the same population. Form 477 deployment records confirm this: near-universal 25/3 Mbps availability and near-universal 10/1 Mbps availability coincide at the tract level. Any residual bias is bounded and benign—since 10/1 Mbps adoption weakly exceeds 25/3 Mbps adoption, the inside-option share is at most slightly overstated, uniformly across tracts, leaving relative demand patterns intact. Table 3 confirms that the implied subscriber count aligns with the FCC national benchmark.

[11] State allocation totals available at: `https://www.ntia.gov/funding-programs/internet-all/broadband-equity-access-and-deployment-bead-program/program-documentation/state-allocation-totals`.

[12] Provider names in Form 477 contain substantial variation across subsidiaries and legacy brands. I consolidate these into canonical corporate-family identifiers; details are in Appendix C.

2.3 Sample Construction

The structural baseline is dated December 2020, before ACP and before any BEAD-induced deployment, so that simulated discounts and fixed-cost reductions apply to a market untouched by either program. Plan prices for observed products and feasible unobserved positions come from the FCC Urban Rate Survey (URS), one of the few nationally representative sources of plan-level broadband prices and now a standard input to work on broadband competition, adoption, and infrastructure policy (e.g., Espín and Rojas, 2024, Kearns, 2024). I match URS plans to tract-level provider-tier products through a six-level brand-first hierarchy prioritizing state, provider, and speed tier, which imputes a price for all 673,110 potential positions. Prices are deflated to constant 2025 dollars by CPI-U. Appendix D gives the full matching algorithm and the distribution of match quality.

2.4 Descriptive Statistics

The estimation sample, summarized in Table 1, depicts a market that is expensive, thinly competitive, and heavily skewed across speed tiers. The typical plan costs \$71.65 per month (median \$69.99). Speeds are sharply right-skewed, so the median product is a better reference point than the mean. The average tract is served by just 2.36 providers offering 3.20 products, confirming the limited choice documented elsewhere [Flamm and Varas, 2022], and the median tract has 1,561 households, median income of \$59,904, and a 70 percent adoption midpoint. Market-share statistics appear in Table 2, once the share-construction procedure is in place.

2.5 Market Share Construction

Estimating broadband demand confronts one binding data limitation: product-level market shares are not observed. The FCC discloses where providers operate, their technologies, and their advertised speeds, but never the subscriber count for a given provider-speed product in a given tract—the exact object the demand system requires.

I recover these shares by combining observed adoption totals with product availability, following the logic of Crawford et al. [2019]. Two steps suffice. FCC tract-level adoption data, reported in binned ranges for any fixed broadband and for service at or above 10/1 Mbps, anchor total adoption and the outside option. Logit weights built from product characteristics and demographics then allocate that total across the available provider-speed-tier products. The resulting shares are inferred, not observed subscriber counts, and should be read as such. *Implementation details, the allocation equation, and validation diagnostics are in Appendix E.*

Table 2 reports summary statistics for the constructed market shares.

Table 1: Descriptive Statistics: Structural Estimation Sample

Variable	Mean	Std. Dev.	P25	Median	P75
Product-level variables					
Price ($/month)	71.65	25.34	49.99	69.99	86.17
Download speed (Gbps)	0.337	0.713	0.025	0.100	0.300
Upload speed (Gbps)	0.065	0.136	0.003	0.020	0.030
High-speed tier ($\geq$25 Mbps/$\geq$3 Mbps)	0.624	–	–	–	–
Tract-level variables					
Products per tract	3.20	1.46	2	3	4
Providers per tract	2.36	1.19	2	2	3
Households	1,679	820	1,126	1,561	2,089
Population	4,511	2,325	2,967	4,169	5,601
Median household income ($)	67,152	33,805	44,318	59,904	82,159
Renter share	0.364	0.228	0.183	0.314	0.509
College share	0.311	0.193	0.161	0.261	0.425
Age 65+ share	0.164	0.080	0.114	0.157	0.200
Broadband adoption share	0.668	0.171	0.559	0.700	0.803

Notes: Product-level variables are computed over the 220,821 observed tract-provider-tier products used in demand estimation. Tract-level variables are computed once per census tract and are not product-weighted. The larger entry sample contains 673,110 potential product positions, which include feasible but unobserved county-footprint positions. The high-speed tier is defined by the federal broadband standard: download speed of at least 25 Mbps and upload speed of at least 3 Mbps. Market-share statistics are in Table 2.

Table 2: Market Share Summary Statistics

Variable	Mean	Std. Dev.	P25	P75
Product market share $\hat{s}_{jt}$	0.259	0.152	0.159	0.324
Within-tier share $\hat{s}_{j\|g,t}$	0.313	0.166	0.202	0.383
Outside-option share s_{0t}	0.171	0.134	0.100	0.300
Provider Herfindahl–Hirschman Index	5,712	2,584	3,694	6,394

Notes: Product shares are inferred using FCC adoption information and product-level data (see Section 2.5). The Herfindahl–Hirschman Index is computed within tracts using provider shares among fixed-broadband subscribers and is scaled from 0 to 10,000. The sample contains 220,821 observed provider-tier products in 69,085 census tracts.

The constructed sample reproduces national aggregates closely enough to serve as a credible counterfactual baseline. Table 3 benchmarks it against FCC totals: the 69,085 tracts cover 116.0 million households (94 percent of the national benchmark), and the inferred adoption shares imply 92.7 million residential fixed connections at 10/1 Mbps or above, within two percent of the FCC figure. The median tract adoption ratio also tracks the benchmark.

Appendix E reports the model-implied allocation of subscribers across providers as a further check on the share-construction procedure.

Table 3: Aggregate Validation of the Structural Sample

Moment	Structural sample	FCC national benchmark	Coverage
Households	116.02 million	123.67 million	93.82%
Residential fixed connections, ≥10/1 Mbps	92.75 million	94.45 million	98.20%
Median tract adoption ratio	0.90	0.89	–

Notes: The structural sample contains 69,085 census tracts. Residential fixed connections are computed as tract households multiplied by the FCC-based 10/1 adoption-share bin. Product speed tiers in the structural model remain defined using advertised 25/3 Mbps availability. FCC national benchmarks are from Federal Communications Commission [2022]. Coverage is the structural-sample value divided by the FCC national benchmark. These moments validate the scale of the structural sample against national residential broadband aggregates; they are not provider-level subscriber counts.

Broadband market structure. How many providers serve a tract governs both the competitive pressure on price and the room for subsidy-induced entry, so the distribution of provider counts is a primitive of the analysis. Table 4 reports it.

Table 4: Provider Competition in the Sample

Number of providers in tract	Share of tracts (%)	Share of households (%)
One provider	23.10	21.96
Two providers	40.16	39.22
Three or more providers	36.73	38.82

Notes: Provider counts based on canonical providers with positive constructed product shares in the tract. Sample: 69,085 census tracts in December 2020.

The picture that emerges is one in which competition is the exception rather than the rule. Nearly a quarter of tracts have exactly one fixed-broadband provider; nearly two-thirds have at most two. The 21.96 percent of households in monopoly tracts face no intra-market price competition at all, and only 36.73 percent of tracts reach three providers or more. This thin competitive landscape is itself part of what the federal programs are designed to address—and it is precisely what gives subsidy-induced entry room to move consumer welfare.

3 Reduced-Form Evidence

3.1 State-Year Panel and Treatment Variables

The event study traces how broadband prices, product menus, and concentration moved in states more exposed to each program. The panel is balanced over 52 states (including Puerto Rico and the District of Columbia) from 2014 to 2025, yielding $N = 624$ state-year cells before missing values. Outcomes are built from FCC Urban Rate Survey plan records, aggregated to state-year plan-count-weighted average prices and plan-composition shares and deflated to 2025 USD by the CPI-U. Six outcomes span three economic margins: prices (log average price and log prices by speed tier), product mix (plan shares below and above the 25/3 Mbps threshold), and concentration (provider HHI). Table 5 reports descriptive statistics.

Table 5: Descriptive Statistics – Main Outcomes

Outcome	N	Mean	SD	Min	Median	Max
Prices						
Log avg. real price	597	4.583	0.190	3.827	4.556	5.254
Log price (<25/3 Mbps)	563	4.200	0.210	3.314	4.223	5.073
Log price (≥25/3 Mbps)	587	4.703	0.223	4.169	4.673	5.483
Plan Composition						
Plan share (<25/3 Mbps)	597	0.261	0.260	0.000	0.171	1.000
Plan share (≥25/3 Mbps)	597	0.739	0.260	0.000	0.829	1.000
Market Structure						
Provider HHI	597	0.524	0.238	0.146	0.477	1.000

Notes: Unit of observation is state × year, 2014–2025 ($N = 624$ state-year cells before missing). These six outcomes are the main outcomes in the event studies and difference-in-differences regressions. Prices are weighted-average plan prices deflated to 2025 USD using the Consumer Price Index for Urban Consumers. Plan shares and the HHI are bounded $[0, 1]$.

Panel structure and outcome construction. The window straddles both rollouts with room to test pre-trends. ACP has eight pre-period years (2014–2021), three program years (2022–2024), and one follow-up year (2025); BEAD has seven pre-announcement years (2014–2020), the enactment year (2021), and four post-enactment years (2022–2025). Five pre-period leads give the design power against parallel-trends violations, and four to five post-period years are enough to trace dynamic effects.

One feature of the data tilts the BEAD estimates toward zero. The Urban Rate Survey samples urban tracts, so the panel captures urban and suburban conditions and underrepresents the rural markets BEAD is built to serve. If BEAD's price and concentration effects concentrate in rural areas—as the structural results below imply—the URS-based estimates understate its true impact, making them a conservative lower bound. ACP, which works through demand expansion in already-served and largely urban markets, sits squarely within the URS frame and is measured more directly.

Both treatment intensities are predetermined and time-invariant, which removes the endogeneity that plagues contemporaneously measured treatment. The ACP eligibility share ($\mathrm{FPL200}_i$) is taken from the 2020 ACS, before launch; the BEAD intensity ($\log \mathrm{BEAD}_i$) is fixed by a federal formula over pre-grant unserved-location counts. Neither responds to ISP behavior or state outcomes over the window, supporting the assumption that exposure is uncorrelated with differential trends.

ACP exposure. I measure state-level ACP eligibility exposure as the share of households with incomes below 200% of the federal poverty level ($\mathrm{FPL200}_i$), drawn from the 2020 American Community Survey. This is a time-invariant, pre-determined cross-sectional measure that varies across states and proxies for the potential demand stimulus ACP can generate. The program became active in 2022; I set $\mathrm{Post}_t^{\mathrm{ACP}} = \mathbf{1}[t \geq 2022]$.

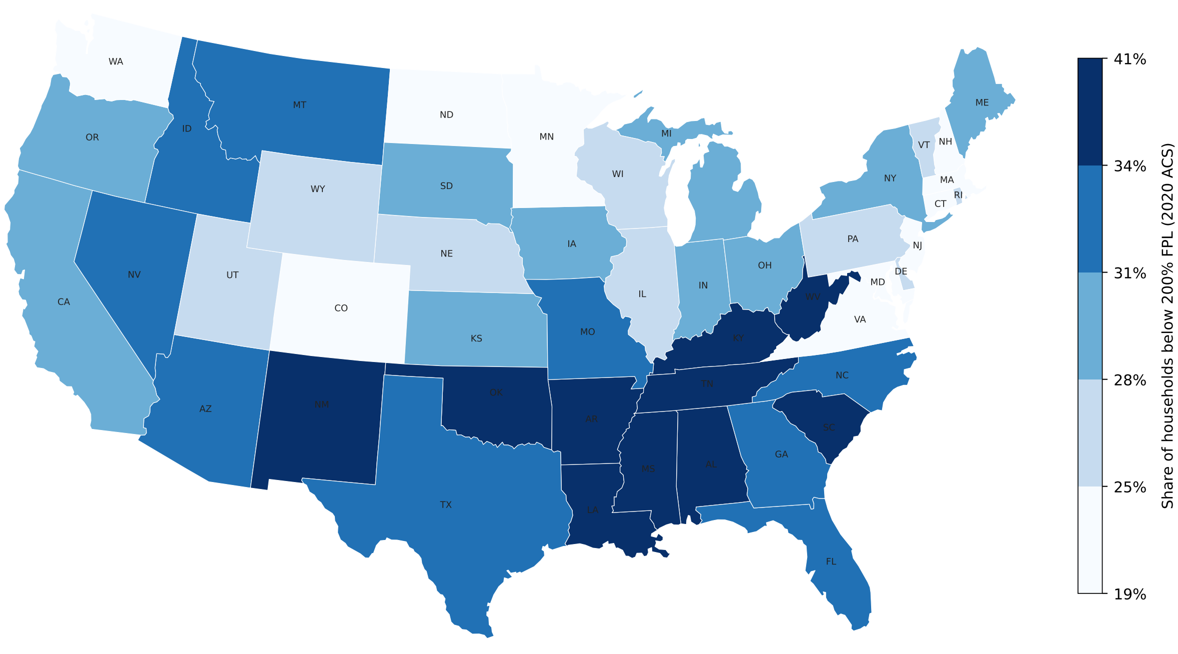

Figure 1: State-level ACP exposure: share of households with incomes below 200% of the federal poverty level ($\mathrm{FPL200}_i$), measured from the 2020 American Community Survey. Darker shading indicates higher eligibility exposure. Colour breaks at sample quintiles.

BEAD exposure. I measure state-level BEAD intensity as the log of BEAD allocation (USD) per unserved broadband location ($\log \text{BEAD}_i$). A higher value indicates that a state receives more funding per targeted location, implying a more intensive supply-side intervention. This measure is also time-invariant and pre-determined by federal allocation formulas; states cannot influence their allocation. BEAD was announced in 2023; I set $\text{Post}_t^{\text{BEAD}} = \mathbf{1}[t \geq 2023]$.

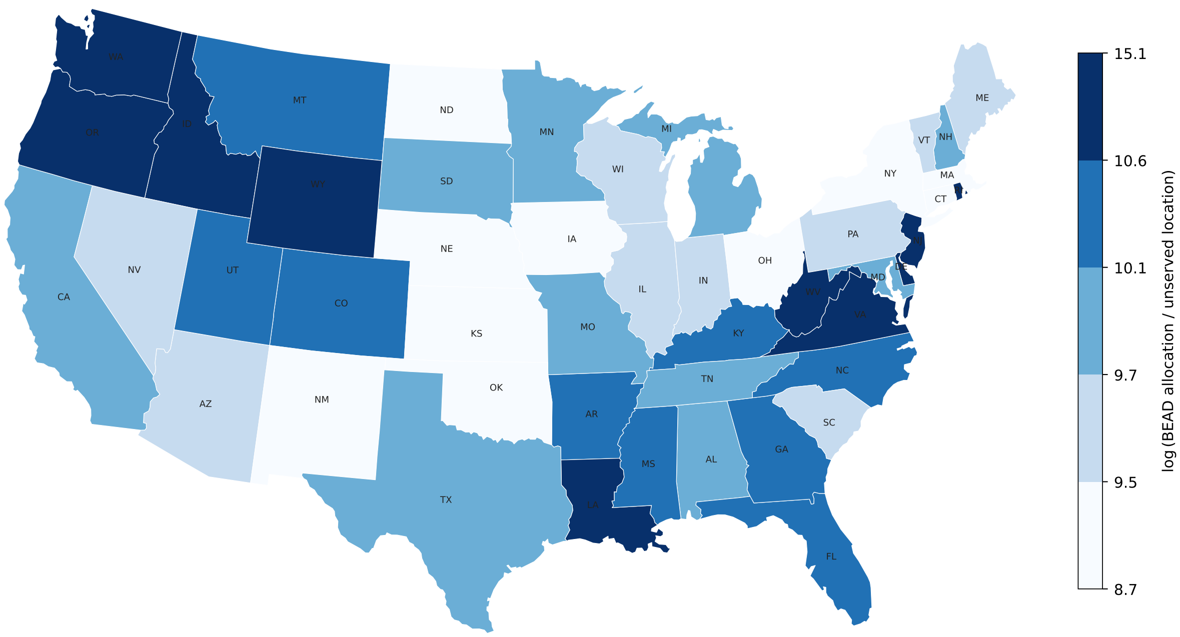

Figure 2: State-level BEAD exposure: log of BEAD allocation (USD) per unserved broadband location ($\log \text{BEAD}_i$), based on NTIA allocation announcements (June 2023). Darker shading indicates higher funding intensity per targeted location. Colour breaks at sample quintiles.

3.2 Identification Strategy

For program $P \in \{\text{ACP}, \text{BEAD}\}$, let D_i^P denote the time-invariant treatment intensity for state i and let τ index event time relative to the policy year. The dynamic specification is:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{\substack{\tau=-5 \\ \tau \neq -1}}^{\bar{\tau}^P} \beta_\tau^P \Big(D_i^P \times \mathbf{1}[t = \bar{t}^P + \tau] \Big) + \boldsymbol{\gamma}_i' \, t + \varepsilon_{it}, \tag{3.1}$$

where α_i are state fixed effects, λ_t are year fixed effects, $\bar{t}^{\text{ACP}} = 2022$ and $\bar{t}^{\text{BEAD}} = 2021$. The omitted period is $\tau = -1$ (the year before the policy), so all estimates are relative to that baseline. Regressions are weighted by the number of plans n_{it} and standard errors are clustered at the state level.

The role of fixed effects. State fixed effects α_i absorb all time-invariant determinants of broadband outcomes at the state level, including permanent differences in geographic coverage, regulatory environment, and demographic composition. Identification therefore exploits *within-state* variation in outcomes over time, purged of any level differences across states. Year fixed effects λ_t absorb common national shocks to broadband markets, such as economy-wide shifts in input costs, federal regulatory changes, or aggregate demand trends. The interaction of treatment intensity with year indicators then captures whether states more intensively exposed to each program moved differentially relative to the national trend.

State-specific trends. For both ACP and BEAD, I include state-specific linear time trends $\boldsymbol{\gamma}_i' t$ to account for the possibility that treatment intensity correlates with pre-existing state-level broadband trajectories. For ACP, states with higher poverty rates—which mechanically have higher eligibility exposure—may have been following divergent broadband price and adoption paths before the program launched. For BEAD, states with more unserved locations tend to be lower-density and lower-income, and may have been experiencing systematically different infrastructure investment trends prior to IIJA enactment. The digital divide literature has documented that broadband adoption and pricing trends differ systematically across income and demographic groups [Goldfarb and Prince, 2008, Espín and Rojas, 2024]. Without state-specific trends, such pre-existing divergence would contaminate both sets of estimates. State-specific trends absorb smooth state-level deviations from the national year effects; the treatment coefficients $\hat{\beta}_\tau^{\text{ACP}}$ and $\hat{\beta}_\tau^{\text{BEAD}}$ are then identified from non-linear departures from each state's pre-existing trend coinciding with program rollout.

Assumption 1 (Conditional parallel trends)**.** *Conditional on state fixed effects, year fixed effects, and state-specific linear time trends, the untreated potential outcome follows the same trend across states with different treatment intensities D_i^P. Formally, $\mathbb{E}[\varepsilon_{it} \mid D_i^P, i, t] = 0$ for all $\tau < 0$.*

The identifying assumption is that states with higher treatment intensity D_i^P were not on systematically different trajectories for reasons unrelated to the policy. Because both FPL200_i and $\log \text{BEAD}_i$ are predetermined by pre-program demographic composition and federal allocation formulas, the main threat to identification would be that states with higher exposure were on different broadband trends before the programs launched. The pre-period coefficients $\hat{\beta}_\tau^P$ for $\tau < 0$ serve as direct falsification tests of Assumption 1.

A secondary concern is the anticipation of program benefits: ISPs or households might alter behavior before formal program launch in expectation of the policy. Under the IIJA timeline, BEAD was announced in November 2021 but allocations were not finalized until 2023, so anticipatory responses would appear as effects at $\tau = 1$–2 (2022–2023), which is largely consistent with what I find (early effects are near zero). For the Affordable Connectivity Program, its predecessor (the Emergency Broadband Benefit program) ran from February to December 2021, meaning some anticipatory ISP behavior may already be absorbed in the $\tau = -1$ base period; this would lead me to understate ACP's total effect. Neither concern appears to drive the main qualitative results.

3.3 ACP: Event Study Results

Figure 3 presents event study estimates from (3.1) for six key outcomes. Event time is indexed relative to 2022 ($\tau = 0$); the dashed vertical line marks November 2021 (IIJA enactment). The omitted period is $\tau = -1$ (year 2021).

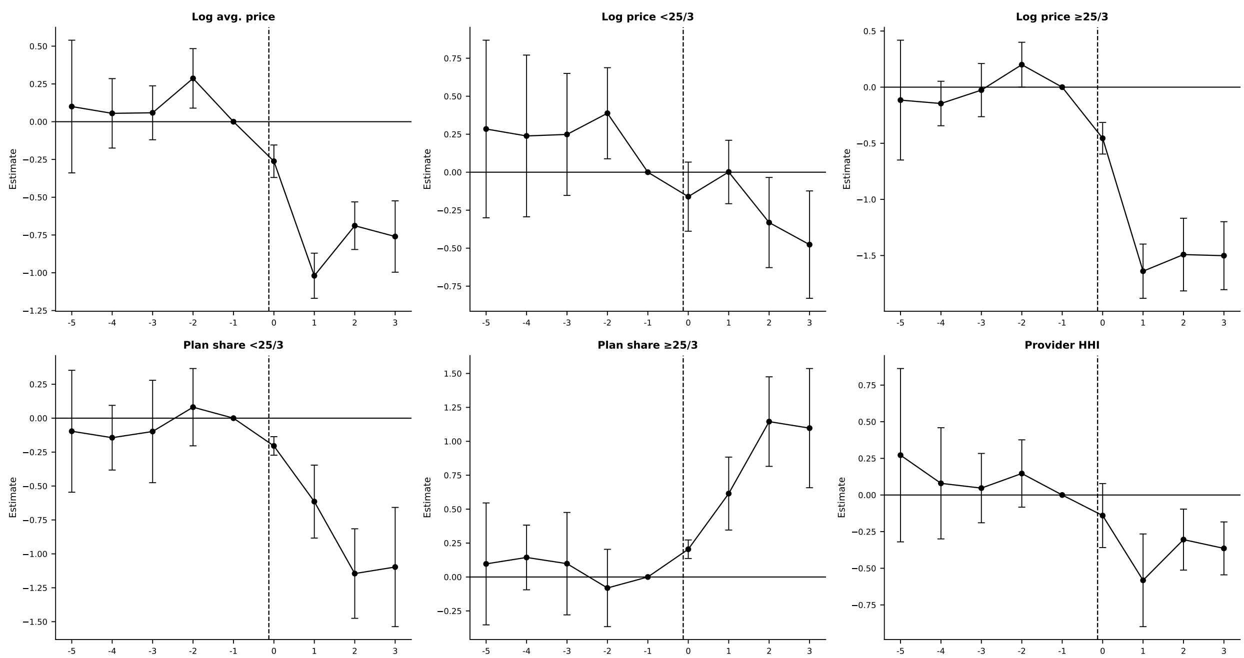

Figure 3: ACP event study estimates. Treatment intensity: share of households below 200% FPL (FPL200_i). TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Event time $\tau = 0$ corresponds to 2022 (ACP launch); the dashed vertical line marks $\tau = 0$. The omitted period is $\tau = -1$ (2021). Bars indicate 95% confidence intervals.

Pre-trends. The pre-period pattern is broadly consistent with the parallel-trends assumption, conditional on state and year fixed effects and state-specific linear trends. For most outcomes (log prices for high-speed plans, plan shares, and provider HHI), the pre-period coefficients are small and close to zero. Log price for plans below 25/3 Mbps shows somewhat elevated pre-period point estimates at $\tau = -5$ to $\tau = -3$, though with wide confidence intervals. This motivates the inclusion of state-specific linear trends, which absorb differential pre-existing trajectories in broadband prices across states at different poverty rates.

Post-ACP effects. ACP moved prices, not just subscriptions. After the 2022 launch, average broadband prices fall in higher-eligibility states: the log-average-price coefficient $\hat{\beta}_1^{\text{ACP}}$ is roughly -1.0 per unit of FPL200 share. With an interquartile range in state eligibility of about 0.10 (0.27 at P25, 0.37 at P75), this means a high-eligibility state saw prices fall about 10 percent more than a low-eligibility state at the peak ($\tau = 1$), settling near 7 percent through $\tau = 2$–3. The effect is largest for high-speed plans, where the response of -0.15 to -0.18 implies a 15–18 percent interquartile gap that persists through $\tau = 2$–3.

The product mix shifts in tandem: the low-speed plan share contracts and the high-speed share expands through $\tau = 2$–3, the pattern expected when subsidized households raise demand for faster service and ISPs upgrade their menus to meet it. Provider HHI also declines in more exposed states, though imprecisely.

Taken together, these are equilibrium responses, not a mechanical subscription bump. Prices fell and menus tilted toward faster plans in the states where ACP made the most households eligible—signatures of an expanded customer base rather than a simple transfer to existing subscribers. The reduced form cannot disentangle whether the price movement reflects pass-through, strategic repricing, or composition; the structural model below isolates the demand-expansion channel that drives it.

3.4 BEAD: Event Study Results

Figure 4 presents event study estimates for BEAD. Event time is indexed relative to 2021 (IIJA enactment, $\tau = 0$); the omitted period is $\tau = -1$ (year 2020). The specification follows (3.1) with state and year fixed effects, state-specific linear time trends, plan-count weights, and state-clustered standard errors, as described in Section 3.

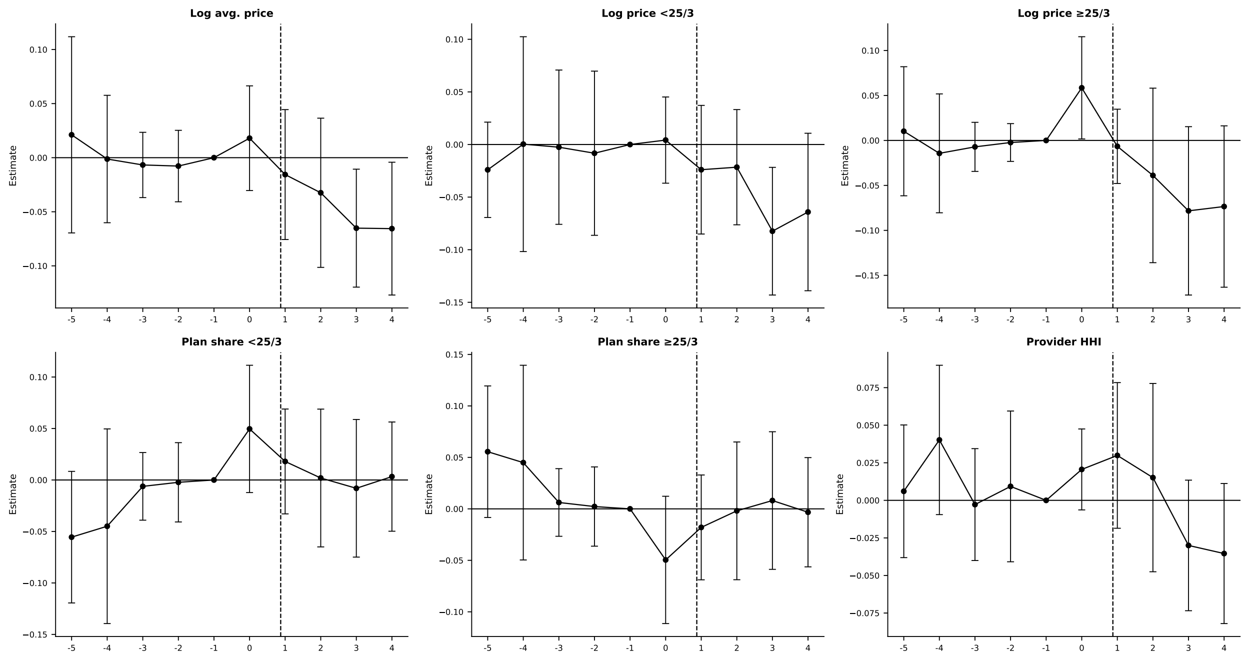

Figure 4: BEAD event study estimates. Treatment intensity: log BEAD allocation per unserved location ($\log \text{BEAD}_i$). TWFE with state-specific linear trends, weighted by number of plans, standard errors clustered by state. Event time $\tau = 0$ corresponds to 2021 (IIJA enactment); the dashed vertical line marks $\tau = 0$. The omitted period is $\tau = -1$ (2020). Bars indicate 95% confidence intervals.

Pre-trends. Pre-period coefficients are small and statistically indistinguishable from zero across all outcomes. States that ultimately received more BEAD funding per unserved location were on similar trajectories in prices, technology, and market structure before the program was announced.

Post-BEAD effects. BEAD's effects are smaller and arrive late, exactly as a deployment subsidy should. Through the first two post-enactment years ($\tau = 1$–2, 2022–2023), every outcome is indistinguishable from zero. Prices begin to move only at $\tau = 3$–4 (2024–2025), falling roughly 0.05–0.07 log points per log unit of BEAD intensity, and provider HHI turns negative over the same window after being flat or slightly positive earlier. Plan composition never responds: high- and low-speed shares stay near zero throughout. This timing is the institutional signature of an entry channel rather than a repositioning channel—networks must be built, interconnected, and activated before new entrants can discipline incumbent prices, and the event window captures only the opening phase of that process.

3.5 Summary Difference-in-Differences Estimates

Pooling the post-period into static two-way fixed-effects difference-in-differences estimates (Table F.1, Appendix F) summarizes the average effect and checks the event-study dynamics: genuine dynamic patterns should reappear as significant pooled coefficients of the same sign.

They do. ACP shows significantly lower prices in higher-eligibility states (log average price −0.561, log high-speed price −0.922), a higher high-speed plan share (+0.390), and lower HHI (−0.306)—the same price and product-mix adjustments the event study traces. BEAD's six estimates are all small and insignificant, consistent with effects that emerge only in the later years and are muted by the urban URS frame. The flat plan-composition estimates reinforce the central reading: BEAD works through entry, not repositioning of existing menus.

3.6 What the Reduced-Form Evidence Establishes

The event studies document the direction, timing, and reduced-form magnitude of each program's effects, but three limitations define the scope of the structural analysis that follows. First, the observed price declines are consistent with multiple channels—direct pass-through, strategic pricing, or changes in plan composition—while the structural model quantifies demand expansion while holding gross provider prices fixed; the two exercises therefore answer different questions. Second, state-level estimates capture average effects across heterogeneous local markets, whereas the structural model exploits rich tract-level variation in market structure, provider presence, and cost conditions that the state panel cannot capture. Third, the event study cannot evaluate alternative policy designs, decompose welfare into consumer and producer surplus, or identify the entry and product-choice responses driving the observed price and concentration patterns. Standard evaluations that count new subscribers and dollars per connection miss these equilibrium responses [Goolsbee and Guryan, 2006]; the structural model makes them central.[13] The reduced-form and structural analyses are therefore complements: the former documents that the programs changed market outcomes; the latter quantifies the channels the model is built to capture and enables counterfactual policy comparisons.

[13] Because ACP was introduced nationally at a common date and exposure varies only through time-invariant eligibility shares, the TWFE estimator is not subject to staggered-adoption bias [Sun and Abraham, 2021, Goldsmith-Pinkham et al., 2020]; the coefficients trace differential dynamic responses to continuous treatment intensity. Appendix F reports static DiD estimates across additional specifications; the qualitative conclusions are stable.

4 Structural Model

4.1 Model Setup and Timing

Markets and products. Markets are indexed by t and firms (or providers) by f. A *market* is a U.S. census tract in 2020, before the implementation of the major IIJA broadband programs.[14] A *product position* is a provider-speed-tier pair (f, g, t), where $g \in \{H, L\}$ denotes the IIJA high-speed tier (≥25/3 Mbps) and low-speed tier (<25/3 Mbps). The 25/3 classification captures a coarse but economically meaningful distinction in service quality relevant for household adoption decisions, distinguishing connections that can reliably support high-bandwidth activities from more limited service options.

I distinguish three product sets. The observed set $\mathcal{J}_t^{obs}$ contains provider-tier products actually offered in tract t; these products enter the demand estimation and markup recovery. The potential set $\mathcal{J}_t^0$ contains provider-tier positions that are feasible in tract t. Feasible but unobserved positions are elements of $\mathcal{J}_t^0 \setminus \mathcal{J}_t^{obs}$; they do not enter the demand estimation as active products, but they are needed for entry and fixed-cost identification. The indicator Y_{fgt} equals one if provider f offers tier g in tract t and zero otherwise, and $Y_t = \{Y_{fgt}\}_{(f,g)\in\mathcal{J}_t^0}$ summarizes the product configuration. Provider entry in this model is *product positioning*: a provider that already serves some tracts in a county expands its product line by activating a tier in an additional tract or offering a second speed tier in an existing tract. The model thus captures both geographic expansion and product-line decisions within the local market.

The *outside option* is no fixed-broadband subscription. Market size M_t is the number of households in the tract, and the market share of the outside option is $s_{0t} = 1 - \sum_{j\in\mathcal{J}_t^{obs}} s_{jt}$.

Timing and information. The following two assumptions formalize the timing and the information structure.[15]

Assumption 2 (Timing)**.** *The game proceeds in two stages:*

1. *Product-positioning stage. Firms observe demand shifters X_t, cost shifters W_t, rivals' observed footprints, and the fixed-cost shocks ζ_{fgt} for all firms. Based on this information, each firm simultaneously chooses its portfolio $Y_{ft} \in \mathcal{Y}_{ft}$.*

[14] I use census-tract data and pre-2021 FCC Form 477 deployment data to ensure the baseline predates IIJA programs. This timing is important for the counterfactuals: simulated price discounts and fixed-cost reductions are applied to markets that do not already reflect ACP or BEAD effects.

[15] The timing assumption has been used in several studies. Examples include Fan and Yang [2024] for the brewery industry, Eizenberg [2014] in the personal computer industry, and Hidalgo [2024] for the broadband industry.

2. *Pricing stage. Conditional on the realized product configuration Y_t, firms observe demand shocks ξ_{jt} and marginal-cost shocks ω_{jt} and simultaneously set monthly plan prices in a complete-information Bertrand-Nash game.*

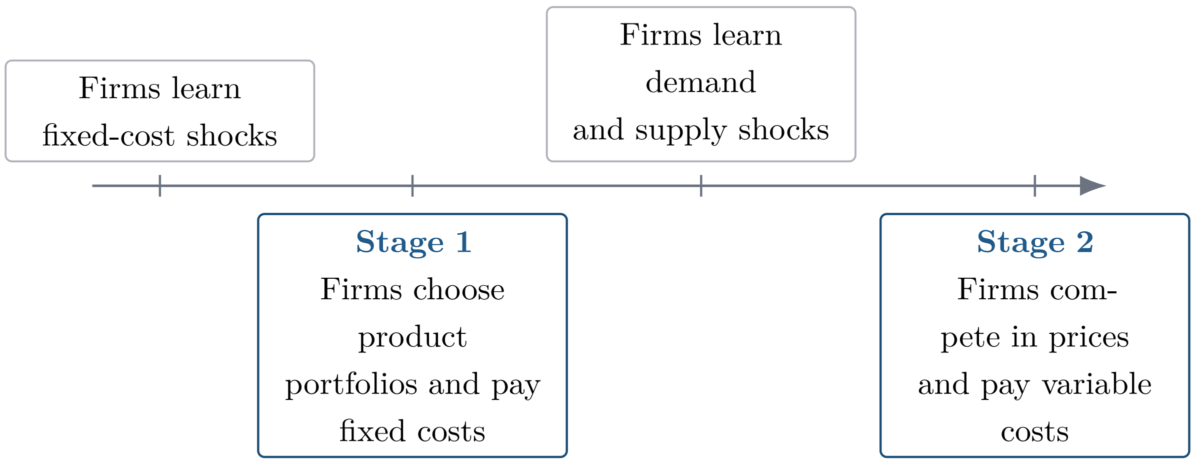

Figure 5: Timing of the structural entry and pricing game.

Assumption 3 (Information)**.** *Firms. At the product-positioning stage, fixed-cost shocks ζ_{fgt}, market characteristics (X_t, W_t), and rivals' potential market footprints are all common knowledge. The demand and cost shocks (ξ, ω) that will be realized in the second-stage pricing game are not yet known; only their joint distribution is common knowledge. Firms therefore compute expected variable profits by integrating over (ξ, ω).*

Econometrician. The shocks (ξ, ω, ζ) are unobserved. Market characteristics (X_t, W_t) and the set of observed product choices $\{Y_{fgt}\}$ are observed.

The separation between portfolio choice and pricing is standard in the entry and product-positioning literature [Mazzeo, 2002, Seim, 2006, Fan, 2013, Eizenberg, 2014] and reflects the different time horizons involved: infrastructure deployment decisions are made months or years before price lists are set. The complete-information assumption on fixed costs is both analytically convenient and economically natural, as ISP deployment costs depend on local permitting, right-of-way negotiations, and contractor availability that are broadly understood within local infrastructure markets. Likewise, price equilibria are modeled under complete information because prices are publicly advertised and routinely monitored by competing firms.

Firm f's first-stage objective is

$$\max_{Y_{ft} \in \mathcal{Y}_{ft}} \mathbb{E}_{\xi,\omega} \left[r_t^f \left(Y_{ft}, Y_{-ft} \right) \mid I_{ft} \right] - \sum_{g \in \{H,L\}} Y_{fgt} FC_{fgt}, \tag{4.1}$$

where $Y_{ft} = (Y_{fHt}, Y_{fLt})$, Y_{-ft} denotes rivals' portfolios, and $r_t^f(\cdot)$ is variable profit from the pricing game. Because a provider can offer at most two tiers in a tract, $\mathcal{Y}_{ft}$ contains four portfolios: no product, high-speed only, low-speed only, and both tiers.

The obstacle to estimation is dimensionality. With about ten feasible provider-tier positions per tract, the configuration space Y_t contains hundreds to thousands of portfolios, and solving the entry game inside an estimation loop across 69,085 markets is infeasible. The scalable discrete-game method of Fan and Yang [2024] bypasses this: rather than solving the game, it bounds the incremental profit from each position, and these bounds imply moment inequalities for the fixed-cost parameters. The game is solved only afterward, when counterfactual policies are evaluated by best response under the new environment.

4.2 Consumer Demand

Household h's indirect utility from product j in market t is

$$U_{hjt} = \delta_{jt} + \nu_{hg} + (1-\rho)\varepsilon_{hjt}, \tag{4.2}$$

where δ_{jt} is mean utility, ν_{hg} is a group-level taste shock, and ε_{hjt} is Type-I extreme value.[16] Mean utility is

$$\delta_{jt} = X'_{jt}\beta + \alpha p_{jt} + \xi_{jt}. \tag{4.3}$$

The vector X_{jt} includes download speed, upload speed, their squares, and a high-speed tier indicator. The parameter α is the marginal utility of price and ξ_{jt} is unobserved quality.

All fixed-broadband products sit in a single inside nest and compete against the outside option of no subscription. The single nest is a deliberate tractability choice, defensible on two grounds. It applies uniformly across all 69,085 tracts: a speed-tier nest would be degenerate wherever only one tier is offered—the within-nest share equals one and the structure collapses to logit—and many rural tracts carry only a low-speed tier, so a tier nest is infeasible without dropping or special-casing those markets. It also yields a unique closed-form Bertrand-Nash price equilibrium, which the entry counterfactuals require: equilibrium prices must be recomputed at every iteration of the positioning game across tens of thousands of markets and fixed-cost draws, and richer structures (random coefficients or a second nest) can admit

[16] This paper departs from Fan and Yang [2024], who employ a random-coefficients demand model. Although such models allow for richer substitution patterns, they may generate non-unique pricing equilibria in multiproduct settings [Nocke and Schutz, 2018], thus complicating welfare and entry counterfactuals. Therefore, I use the nested logit model, which delivers a unique closed-form pricing solution and ensures a stable mapping from demand primitives to prices (e.g., Berry [1994]). Tractability is essential for evaluating firms' endogenous product offerings and entry decisions. I nonetheless rely on Fan and Yang [2024]'s framework to recover fixed-cost parameters in the entry stage.

multiple pricing equilibria in multiproduct markets [Nocke and Schutz, 2018]. The single nest is thus chosen for uniqueness and speed, not because tier substitution is believed absent. The Berry inversion [Berry, 1994] gives

$$\log(s_{jt}) - \log(s_{0t}) = X'_{jt}\beta + \alpha p_{jt} + \rho \log(s_{j|g,t}) + \xi_{jt}, \tag{4.4}$$

where $s_{j|g,t} = s_{jt}/(1 - s_{0t})$. In the tract-level data, the left-hand side is the CSS mean utility recovered from product-level implied shares.

Price and the within-inside share are endogenous. I instrument for these variables using BLP-style rivals' characteristics [Berry et al., 1995]:

$$z_{jt}^{(k)} = \frac{1}{J_t - 1} \sum_{\ell \neq j} X_{\ell t}^{(k)}, \qquad k \in \{\text{downstream}, \text{upstream}, \text{Tier H}\}, \tag{4.5}$$

and the number of same-tier rivals [Aguirregabiria et al., 2024].

The logic of these instruments is standard [Berry et al., 1995, Nevo, 2001, Conlon and Gortmaker, 2020]. Rivals' speed characteristics affect a firm's own price only through strategic interaction: a firm with slower competitors can charge higher prices without losing as many subscribers. These characteristics are plausibly exogenous to a given firm's unobserved quality shock ξ_{jt} if firms choose deployment locations for reasons uncorrelated with rivals' cost shocks. The number of same-tier rivals captures the direct competitive pressure within the speed tier, which affects pricing but is not itself a choice variable of the focal firm in any given tract.

The preferred specification absorbs State and Provider × Tier fixed effects. State fixed effects absorb time-invariant state-level quality differences in broadband markets. Provider × Tier fixed effects absorb all provider–tier–specific mean quality differences, eliminating any systematic differences between, say, fiber products versus DSL products or between large national ISPs and small regional ones. Identification of $\hat{\alpha}$ and $\hat{\rho}$ in this specification comes from within-provider-tier variation across tracts in relative prices and market shares, conditional on state-level and provider-tier-level averages.

4.3 Oligopoly Price Competition

In the pricing stage, all product configurations Y_t are realized and common knowledge. Firm f's variable profit is

$$r_t^f = \sum_{j \in \mathcal{J}_t^f} (p_{jt} - mc_{jt})\, M_t\, s_{jt}(p_t). \tag{4.6}$$

Differentiating with respect to p_{jt} and setting to zero gives the Bertrand-Nash first-order condition:

$$s_{jt} + \sum_{k \in \mathcal{J}_t^f} (p_{kt} - mc_{kt}) \frac{\partial s_{kt}}{\partial p_{jt}} = 0. \tag{4.7}$$

Stacking over all products yields the linear system:

$$s_t + (H_t \odot \Omega_t)(p_t - mc_t) = 0, \tag{4.8}$$

where H_t is the $J_t \times J_t$ matrix of demand derivatives with $(H_t)_{jk} = \partial s_{kt}/\partial p_{jt}$, and Ω_t is the ownership matrix with $(\Omega_t)_{jk} = 1$ if j and k are offered by the same firm. The system is solved for the markup vector:

$$\hat{\mu}_t^{BN} = -(H_t \odot \Omega_t)^{-1} s_t, \tag{4.9}$$

and marginal costs are recovered as $\hat{mc}_{jt} = p_{jt} - \hat{\mu}_{jt}^{BN}$. I then estimate the cost regression

$$\log(\hat{mc}_{jt}) = W_{jt}'\gamma + \omega_{jt}, \tag{4.10}$$

where W_{jt} contains speed characteristics and the high-speed tier indicator.

4.4 Product Positioning and Fixed Costs

The supply-side object is a product-positioning decision. A product position is a provider $\times$ speed-tier pair in a census tract. Let f index canonical providers, $g \in \{H, L\}$ index speed tiers, and t index census tracts. The primitive decision is

$$Y_{fgt} = \mathbf{1}\{\text{provider } f \text{ offers speed tier } g \text{ in tract } t\}. \tag{4.11}$$

This definition treats entry as product positioning rather than entry into the national broadband industry. It captures both geographic expansion into a tract and product-line expansion within a tract. For example, a provider that already offers the low-speed tier in tract t enters the high-speed position if it also begins offering the high-speed tier.

Potential product positions. The observed data are organized at the tract $\times$ canonical-provider $\times$ speed-tier level, so each observed row corresponds to $Y_{fgt} = 1$. To estimate fixed costs, I also need feasible positions that are not chosen. I define the potential set using local provider footprints rather than assuming every national provider can enter every tract. Let

$c(t)$ denote the county containing tract t. The baseline potential set is

$$\mathcal{J}_t^0 = \{(f,g) : \exists\, t' \in c(t) \text{ such that } Y_{fgt'} = 1\}\,. \tag{4.12}$$

Thus, provider-tier position (f,g) is feasible in tract t if that same provider-tier pair is observed somewhere else in the county. This restriction is economically natural: broadband expansion depends on nearby network presence, local permitting, and rights of way. It also keeps the potential choice set local enough to avoid treating providers with no plausible footprint as potential entrants.

For every feasible but unobserved position in $\mathcal{J}_t^0$, I impute product characteristics using predetermined information: provider-tier-state means, then provider-tier national means, then tier means. Observed positions retain their observed characteristics.

Table 6 reports the implied dimensions. The average tract contains 3.20 observed products but 10.54 potential provider-tier positions. A full market-level portfolio model would require evaluating $2^{|\mathcal{J}_t^0|}$ configurations per market (up to thousands at the 75th percentile), motivating the Fan and Yang [2024] approach below.

Table 6: Dimension of Observed and Potential Product Sets

Product-set definition	Mean	P25	Median	P75	P90
Observed products in tract, $\mathcal{J}_t^{obs}$	3.20	2	3	4	5
Observed providers in tract	2.36	2	2	3	4
County-footprint positions, $\lvert\mathcal{J}_t^0\rvert$	10.54	6	9	13	21
County providers	8.13	4	7	10	16

Notes: The unit of observation is a census tract. Observed products are provider-tier products present in the tract. The county-footprint potential set includes provider-tier positions observed somewhere in the same county as the tract.

The potential product set is used exclusively for the entry model and fixed-cost identification. Demand estimation and markup recovery use only the observed product set.

4.5 Multiple Equilibria and Equilibrium Selection

The positioning game generically has multiple Nash equilibria: a single fixed-cost draw can support several mutually best-response configurations. This multiplicity, recognized since Bresnahan and Reiss [1991] and Tamer [2003] and confronted by Ciliberto and Tamer [2009] and Fan and Yang [2024], normally obstructs estimation because it severs the map from parameters to a unique prediction.

It does not obstruct estimation here. The moment-inequality approach asks only that the observed configuration be *not dominated*, a condition every Nash equilibrium satisfies, so the fixed-cost bounds remain valid whatever selection rule operates in the data. This is the decisive advantage of the Fan and Yang [2024] method over likelihood approaches that must assume a unique equilibrium.

The counterfactual stage does require solving for post-policy equilibria. I handle this as follows. For each market and each draw of fixed costs, I run iterated best-response from three starting points: the observed pre-policy configuration, the empty configuration, and the full feasible configuration.[17] The algorithm is declared converged when no firm can profitably deviate from the current portfolio, checked by a firm-by-firm Nash verification. If multiple fixed points are found from different starting configurations, I retain all of them. The reported national-level bounds combine equilibrium multiplicity and fixed-cost parameter uncertainty by stacking market-level lower and upper outcomes across equilibria and fixed-cost draws and reporting the 2.5th and 97.5th percentile bounds.

Markets where no fixed point is found within the iteration limit (fewer than one percent of markets) are assigned the pre-policy configuration. This conservative fallback ensures the bounds remain valid; it is likely to understate the response of hard-to-solve markets. Extensive convergence diagnostics, reported in Appendix J, show that the algorithm converges in the vast majority of markets, and the distribution of convergence failures is not systematically correlated with market size or policy intensity.

The main measurement assumptions underlying the structural analysis—inferred product-level market shares, imputed URS prices, and a static December 2020 baseline—are discussed in Appendix G, along with supporting robustness checks.

5 Estimation Results

Estimation proceeds in the order the model is identified. Demand comes first, from the Berry inversion in (4.4), instrumenting price and the within-inside share with rivals' characteristics and the same-tier rival count and absorbing State and Provider $\times$ Tier fixed effects. The demand estimates and the Bertrand-Nash conditions in (4.7) then recover tract-level markups and marginal costs through (4.9). Fixed costs come last, as bounds from the Fan and Yang [2024] positioning inequalities, which require only that observed choices be undominated. Appendix J details the fixed-cost moments, weight matrix, and confidence region.

[17] Fan and Yang [2024] consider two starting points: the empty and the full feasible configuration.

5.1 Consumer Demand

Table 7 reports four demand specifications. Columns (1) and (2) are OLS; columns (3) and (4) instrument for price and the within-inside share. Even-numbered columns absorb State and Provider × Tier fixed effects. Column (4) is the preferred specification.

Table 7: Nested Logit Demand Estimates

	OLS		2SLS (BLP instruments)	
	(1)	(2)	(3)	(4)
Price / \$100 [$\hat{\alpha}$]	-0.260^{***}	-0.329^{***}	0.818^{***}	-9.424^{***}
	(0.002)	(0.004)	(0.005)	(0.696)
Within-share [$\hat{\rho}$]	0.245^{***}	0.214^{***}	0.255^{***}	0.220^{***}
	(0.001)	(0.001)	(0.002)	(0.008)
Download speed (Gbps)	0.308^{***}	0.367^{***}	0.139^{***}	0.435^{***}
	(0.003)	(0.003)	(0.010)	(0.015)
Download speed^2	-0.098^{***}	-0.107^{***}	-0.079^{***}	-0.152^{***}
	(0.002)	(0.002)	(0.009)	(0.004)
Upload speed (Gbps)	0.008^{***}	0.017^{***}	0.248^{***}	0.021^{*}
	(0.003)	(0.003)	(0.009)	(0.011)
Upload speed^2	0.047^{***}	0.051^{***}	0.022^{**}	0.089^{***}
	(0.002)	(0.002)	(0.009)	(0.003)
High-speed indicator	0.122^{***}	n.s.	-0.088^{***}	n.s.
	(0.001)	—	(0.002)	—
N	220,821	220,821	220,821	220,821
R^2	0.782	—	—	—
State FEs	No	Yes	No	Yes
Provider×Tier FEs	No	Yes	No	Yes
Mean own-price elasticity	−0.18	−0.22	0.57	−6.33

Notes: HC1 robust standard errors in parentheses for OLS; heteroskedasticity-robust for 2SLS. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. n.s. = not significant; the high-speed indicator is collinear with Provider×Tier FEs in columns (2) and (4) and is omitted. BLP instruments: mean rivals' downstream speed, upstream speed, tier-H share, and number of same-tier rivals (all tract-level). Column (4) is the preferred specification.

Both endogeneity corrections are needed, and in combination. Instrumenting alone (column 3) is not enough: without fixed effects the price coefficient turns positive (+0.818) and the elasticity wrong-signed, because higher-quality providers charge more and that quality is unobserved. Adding State and Provider×Tier effects in the preferred column (4) absorbs those quality differences, and the price coefficient becomes negative and large—IV raises its magnitude 36× over OLS ($-0.26 \to -9.42$), the expected direction of the bias when price and unobserved quality move together.

The nesting parameter $\hat{\rho} = 0.220$ indicates moderate within-group correlation among fixed-broadband products, significantly below the estimate from the state-level specification ($\hat{\rho} = 0.646$).[18] The difference reflects the different levels of aggregation: at the tract level, product availability varies sharply across providers, reducing within-nest correlation. Download speed enters positively and concavely, consistent with diminishing marginal utility from very high speeds.

The relatively low $\hat{\rho}$ implies substitution patterns that are closer to proportional (multinomial logit) than to perfect within-nest substitution. This is a known limitation of the single-nest specification: with $\hat{\rho}$ near zero, the model does not generate strong cross-tier substitution asymmetries. A two-level nest (speed tier within technology, or technology within speed tier) would allow high-speed and low-speed products to substitute more strongly within tier. As discussed above, however, such a structure is not feasible here because many rural tracts contain only one speed tier, causing the second nest to be degenerate. The tractability requirements of the entry-game counterfactuals further preclude random-coefficients demand, which may generate non-unique pricing equilibria [Nocke and Schutz, 2018]. Given these constraints, $\hat{\rho} = 0.220$ should be interpreted as capturing residual within-broadband correlation after absorbing provider-tier and state fixed effects, not as a claim that all broadband products are equally close substitutes.

The mean own-price elasticity in the preferred specification is -6.33:

$$\varepsilon_{jj} = \frac{\hat{\alpha}}{100} p_{jt} \left[\frac{1}{1-\hat{\rho}} - \frac{\hat{\rho}}{1-\hat{\rho}} s_{j|g,t} - s_{jt} \right]. \tag{5.1}$$

Comparison with related estimates. The mean own-price elasticity of -6.33 exceeds the participation-margin estimates common in the broadband demand literature (-1 to -4, Rosston et al. 2010, Nevo et al. 2016), but is closely aligned with recent product-level broadband estimates: Kearns [2024] reports -4.2 to -4.7, Goetz [2019] finds -5.9, and Crawford et al. [2019] finds -4.1 to -5.4 in cable television. The larger magnitude reflects the product-substitution margin within a local market, not the extensive-margin adoption decision; an alternative state-level aggregation produces a mean elasticity of -5.87, providing reassurance that the tract-level share construction procedure does not mechanically drive the result.[19]

[18] An earlier version of the paper estimated the model at the state level using aggregate broadband adoption shares; the state-level nesting parameter of 0.646 reflects a coarser product mix with less within-state variation in provider availability.

[19] Detailed comparisons across specifications and aggregation approaches are reported in Appendix H.

Elasticity heterogeneity across market types. A key feature of the nested logit model is that own-price elasticities vary with market shares, as seen in (5.1). This heterogeneity has substantive implications for welfare calculations. In low-share markets (typically rural tracts with fewer products and higher outside-option shares), own-price elasticities are smaller in magnitude: at a product share of $s_{jt} = 0.10$ with $\hat{\rho} = 0.220$, the formula yields an elasticity of approximately -3.5. In high-share markets (dense urban tracts with many products and low outside-option shares), own-price elasticities are larger: at $s_{jt} = 0.45$, the elasticity is approximately -8.7. The mean of -6.33 reflects the distribution of market shares across tracts, which is skewed toward intermediate values.

This elasticity heterogeneity matters for the welfare effects of both programs. ACP raises consumer surplus most where elasticity is highest—where consumers are most price-sensitive and closest to the margin of subscribing—which tends to be in lower-income tracts with moderate to high baseline adoption. BEAD, which operates through fixed-cost relief rather than price changes, is less sensitive to demand elasticity and instead depends on the gap between variable profits and fixed costs. The combination of high-elasticity markets (well-served by ACP) and low-elasticity but high-fixed-cost markets (well-served by BEAD) reinforces the geographical complementarity between the two programs documented in Section 6.3.

Instrument strength. Table 8 reports the excluded-instrument coefficients and weak-identification diagnostics for the preferred specification (column 4 of Table 7). Each first-stage regression also includes the full set of included exogenous controls and the fixed effects used in the second-stage demand specification.

With two endogenous regressors, the relevant diagnostics condition on the other endogenous variable. The Sanderson–Windmeijer conditional first-stage statistics are 320.1 for price and 42,082.4 for the within-inside share, both well above conventional weak-instrument thresholds. The Cragg–Donald minimum eigenvalue of 623.2 confirms system-level identification. The signs are economically intuitive: rivals with faster speeds and larger high-speed tier shares face lower own prices (competitive pressure) and lower within-group shares (substitution toward rivals); more same-tier rivals compress both margins. The overidentification test rejects exact exclusion ($\chi^2 = 14.84$, $p = 0.001$), but with more than 216,000 observations the test has high power against small deviations from strict exclusion; I interpret the instruments as strong and approximately valid.

Table 8: First-Stage Diagnostics: Preferred 2SLS-FE Specification

	Price / \$100	$\log(s_{j\|g})$
Mean rivals' download speed (Gbps)	-0.006^{***}	-0.014^{***}
Mean rivals' upload speed (Gbps)	0.004^{***}	0.036^{***}
Mean rivals' high-speed share	-0.006^{***}	-0.276^{***}
Number of same-tier rivals	-0.002^{***}	-0.257^{***}
Included exogenous controls	Yes	Yes
State fixed effects	Yes	Yes
Provider-by-tier fixed effects	Yes	Yes
Angrist–Pischke Wald statistic	435.7	167,617.0
Sanderson–Windmeijer/conditional F	320.1	42,082.4
Interpretation	Strong conditional relevance	Strong conditional relevance
Cragg–Donald minimum eigenvalue statistic	623.2	

Notes: The table reports coefficients on excluded instruments. Each first stage also includes the exogenous controls and fixed effects shown above. The Sanderson–Windmeijer/conditional F statistic tests excluded-instrument relevance for one endogenous variable conditional on the other endogenous variable. Significance: $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

5.2 Marginal Costs and Markups

The preferred supply specification (column 1 of Table 9) excludes fixed effects, because the identifying variation in broadband costs runs across providers, technologies, and speed tiers—dimensions that rich fixed effects absorb. The fixed-effects specification (column 2) changes the signs of several cost shifters to economically counterintuitive directions and increases the share of products with negative implied marginal costs; the parsimonious specification avoids these pathologies.

The preferred specification implies that download speed raises marginal cost at a diminishing rate. High-speed tier products carry a marginal-cost premium of $\exp(0.341) - 1 \approx 41$ percent relative to low-speed tier products. Upload speed is negatively related to marginal cost in the cross-section, consistent with fiber-grade providers combining high upload capacity with newer, more efficient infrastructure. Of 220,821 observed products, 22,025 (9.97 percent) yield negative implied marginal costs, concentrated among multiproduct firms, a known feature of Bertrand-Nash markup recovery in differentiated-products markets [Nevo, 2001].

Conditional on positive marginal costs, the median Lerner index is 18.2 percent. This

Table 9: Supply-Side OLS: Dependent Variable $\log(\hat{mc}_{jt})$

	(1) No FE Preferred	(2) Two-way FE
Constant	3.742^{***}	—
	(0.002)	
Download speed (Gbps)	0.287^{***}	-0.022^{***}
	(0.010)	(0.003)
Download speed^2 (Gbps^2)	-0.026^{***}	-0.003^{***}
	(0.009)	(0.001)
Upload speed (Gbps)	-0.412^{***}	-0.003
	(0.009)	(0.003)
Upload speed^2 (Gbps^2)	0.039^{***}	0.006^{***}
	(0.009)	(0.001)
High-speed indicator	0.341^{***}	—
	(0.003)	
Observations	198,796	198,796
R^2	0.226	0.001
RMSE	0.394	0.173
State FE	No	Yes
Provider×Tier FE	No	Yes

Notes: Dependent variable is $\ln(\hat{mc}_{jt})$ where $\hat{mc}_{jt} = p_{jt} - \hat{\mu}_{jt}^{BN}$. Sample restricted to products with positive implied marginal costs. HC1 robust standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Tier H and the constant are collinear with Provider×Tier FEs in column (2) and are omitted.

figure implies a price-cost margin of roughly 18 cents per dollar of revenue for the typical broadband product. For single-product firms—for whom the Bertrand-Nash first-order condition simplifies to the inverse-elasticity rule—the mean Lerner index of 15.3 percent is close to $1/|\hat{\varepsilon}| = 15.8$ percent evaluated at the sample-mean own-price elasticity of -6.33, providing an approximate internal consistency check on the structural parameters.[20] The higher Lerner indices for multi-product firms (mean 27.9 percent) reflect the internalization of cross-product cannibalization effects [Nevo, 2001].

Comparison with related estimates. The median Lerner index of 18.2 percent is plausible relative to estimates in related markets. Nevo [2001] finds Lerner indices of 16–45

[20] The comparison is approximate: the inverse-elasticity benchmark uses the all-product mean elasticity, while the single-product mean Lerner uses only single-product-firm observations. A tighter check would use the single-product-firm mean elasticity directly.

percent for ready-to-eat cereals, a market characterized by high product differentiation and few national competitors. Crawford, Shcherbakov, and Shum [2019] find similar magnitude markups in cable television. The recovery of markups depends on the demand elasticity, and the 18.2 percent median is directly consistent with a mean own-price elasticity of -6.33 through the Lerner formula. Average marginal costs of approximately \$49 per month for all products are economically plausible given the capital intensity of broadband infrastructure. The constant of 3.742 in the log-marginal-cost regression implies $\exp(3.742) \approx \$42$ per month at zero speeds, with speed characteristics adding to this baseline; high-speed products average approximately \$57 per month in marginal costs.

Tables 10 and 11 report the full distribution of markups, marginal costs, and Lerner indices by speed tier and firm type.

Table 10: Bertrand-Nash Markup Distribution by Speed Tier

	All	Tier H	Tier L	$\hat{mc} > 0$ only
Observations	220,821	138,460	82,361	198,796
% negative $\hat{mc}$	9.97%	8.62%	12.25%	—
Panel A: Markup $\hat{\mu}$ (\$/month)				
Mean	23.03	21.62	25.40	—
Median	12.68	12.25	14.11	—
P90	35.37	35.37	106.11	—
Panel B: Marginal Cost $\hat{mc}$ (\$/month)				
Mean	48.78	57.32	34.41	—
Median	54.51	59.82	47.03	—
P90	83.13	87.73	60.77	—
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$				
Mean	—	0.188	0.262	0.215
Median	—	0.156	0.220	0.182
P90	—	0.302	0.452	0.357

Notes: Tier H: download $\geq$25 Mbps; Tier L: <25 Mbps. Lerner index $= \hat{\mu}/p$; values exceeding 1 arise when implied MC is negative. The "$\hat{mc} > 0$ only" column restricts to the 90.0% of products with non-negative implied marginal costs.

Several patterns merit discussion. High-speed (Tier H) products have marginally lower Lerner indices than low-speed (Tier L) products (median 0.156 vs. 0.220). This is consistent with high-speed products facing greater effective competition: fiber and modern cable products occupy a more competitive landscape than legacy DSL. Tier L products, largely DSL, exhibit higher markups in many markets because they operate as de facto monopolists in low-density areas where no competing technology exists. The substantially elevated Lerner indices for multi-product firms (median 0.248 vs. 0.138 for single-product) reflect the well-understood

Table 11: Markup and Lerner Index by Firm Type

	Single-Product	Multi-Product
Observations	105,597	115,224
% negative $\hat{mc}$	4.76%	14.75%
Panel A: Markup $\hat{\mu}$ (\$/month)		
Mean	15.85	29.61
Median	10.86	16.80
P90	15.15	106.11
Panel B: Marginal Cost $\hat{mc}$ (\$/month)		
Mean	64.51	34.37
Median	64.85	48.77
Panel C: Lerner Index $\hat{L} = \hat{\mu}/p$, $\hat{mc} > 0$ only		
Mean	0.153	0.279
Median	0.138	0.248
P25	0.120	0.190
P75	0.174	0.323
P90	0.219	0.463

Notes: Single-product firms offer one tier in the tract; multi-product firms offer both tiers or operate in multiple tracts. Lerner index restricted to products with $\hat{mc} > 0$.

mechanism in differentiated-products markets: firms with multiple products internalize the cannibalization effect between them, setting higher markups on each product than a single-product firm would [Nevo, 2001]. In broadband markets, this internalization means that an ISP offering both high-speed and low-speed tiers in the same tract prices both higher than it would if the tiers were offered by independent firms, reducing consumer surplus for buyers who would have benefited from intra-firm competition. This finding has a direct counterfactual implication: BEAD-induced entry, which typically introduces a new provider rather than a new product tier from an existing provider, generates competitive pressure that is absent when an incumbent simply adds a second tier.

Figure 6 reports key structural diagnostics. Demand shocks are centered near zero and approximately symmetric, consistent with the orthogonality conditions imposed by the BLP instruments. This is an important internal validity check: if the instruments were correlated with unobserved quality, the residuals would be systematically shifted. Marginal-cost shocks are also approximately symmetric and centered near zero. The markup distribution is right-skewed, reflecting the higher markups of multi-product firms, and is consistent with the Lerner index distributions in Tables 10 and 11 above. Appendix I provides supplementary markup diagnostics including the marginal-cost distribution by technology type.

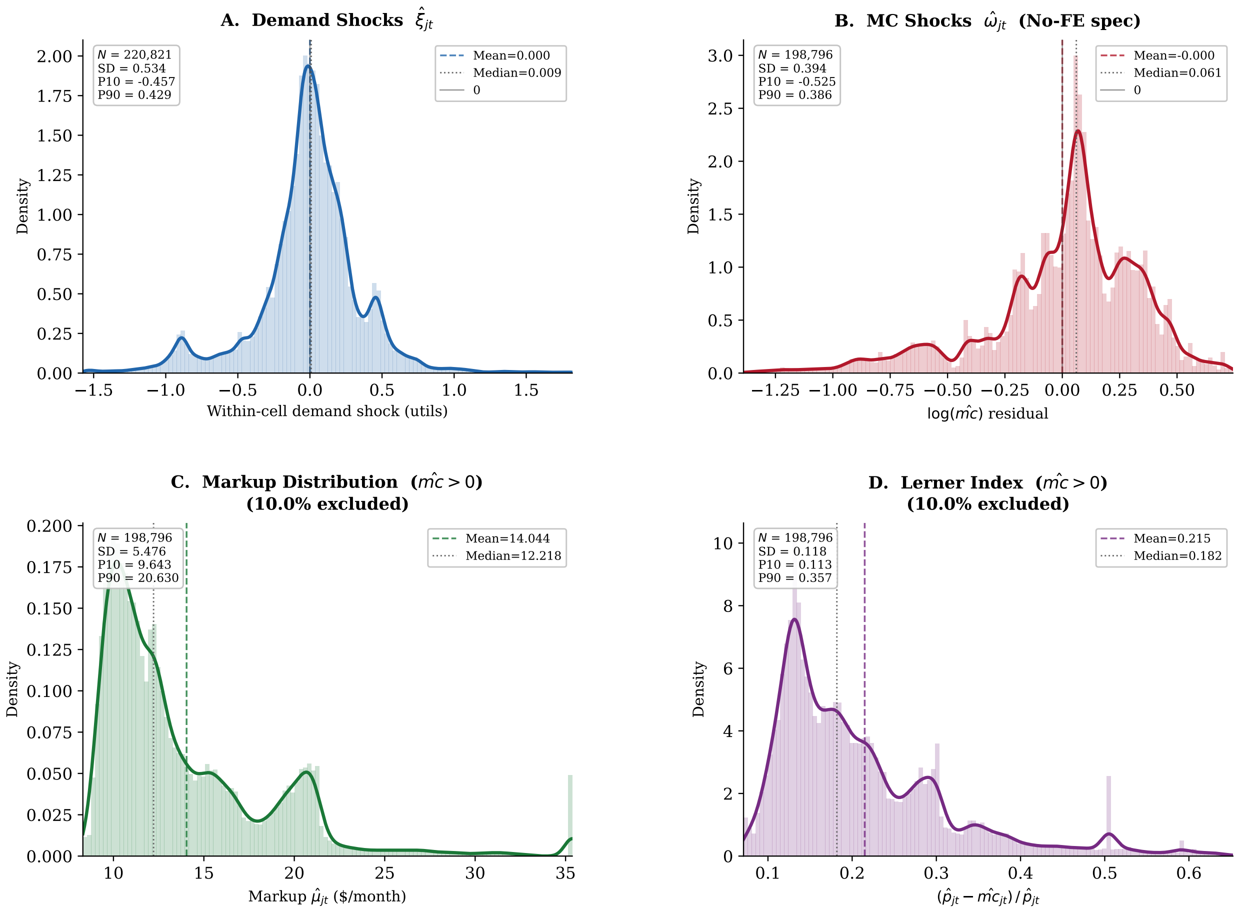

Figure 6: Empirical distributions of demand shocks $\hat{\xi}_{jt}$, marginal-cost shocks $\hat{\omega}_{jt}$, Bertrand-Nash markups, and price-to-marginal-cost ratios, trimmed to the 1st–99th percentile for display. Dashed lines mark means; dotted lines mark medians. The distribution of $\hat{\xi}_{jt}$ is approximately centered at zero and symmetric, consistent with the orthogonality conditions from the demand system. Markups are concentrated in the $10–$30 range with a modest right tail driven by multi-product firms. Price-cost ratios have median near 1.20, reflecting the 18.2 percent median Lerner index.

5.3 Fixed-Cost Estimates

I parameterize fixed costs by market-size cells and speed tier:

$$FC_{fgt} = \theta_{b(t),g} + \sigma_\zeta \zeta_{fgt}, \qquad \zeta_{fgt} \sim N(0,1), \tag{5.2}$$

where $b(t) \in \{\text{Small}, \text{Medium}, \text{Large}\}$ denotes the market-size tertile of tract t and $g \in \{H, L\}$ is the speed tier. This specification allows fixed costs to differ between small, medium, and large markets and between high-speed and low-speed positions, while $\sigma_\zeta \zeta_{fgt}$ captures private position-level cost heterogeneity.

The identification logic is simple. When a provider offers a tier, its fixed cost must be low enough that entry was profitable in at least one equilibrium: the most competitive environment for that position bounds variable profit from below, and the fixed cost cannot exceed that lower bound. When a feasible tier is absent, the fixed cost must be high enough that entry was unprofitable in every equilibrium: the most favorable environment bounds variable profit from above, and the fixed cost must exceed that upper bound. Together, these dominance conditions generate moment inequalities over the six fixed-cost parameters without requiring equilibrium selection [Fan and Yang, 2024, Ciliberto and Tamer, 2009, Tamer, 2003].

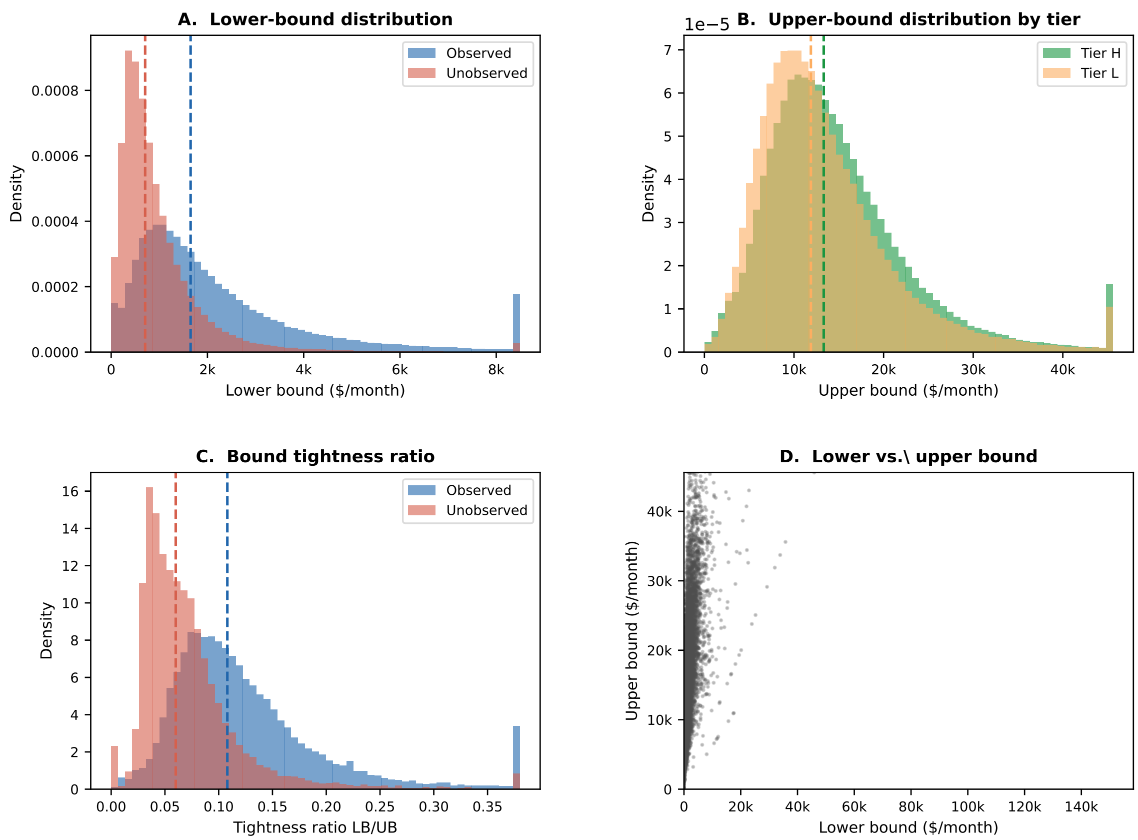

Figure 7: Fan–Yang fixed-cost bounds from product-positioning decisions. Each panel illustrates the dominance restrictions that bind for an observed entry or non-entry: observed products generate upper bounds on fixed costs (the firm's variable profit exceeds the fixed cost); unobserved feasible products generate lower bounds (variable profit does not cover the fixed cost at any equilibrium). The moment inequalities aggregate these restrictions across positions to produce identified bounds on the six fixed-cost parameters.

Figure 7 summarizes the identifying logic. Further sanity checks are available in Appendix K. Table 12 reports the resulting bounds.

Table 12: Fixed-Cost Parameter Bounds by Market Size and Speed Tier

Market size	Tier	Bound (θ)	SD bound (σ_ζ)
Small	High speed	[717, 9,790]	[1,734, 5,068]
	Low speed	[532, 8,572]	[886, 3,864]
Medium	High speed	[1,197, 15,773]	[3,191, 7,875]
	Low speed	[900, 13,862]	[1,578, 5,081]
Large	High speed	[1,930, 24,579]	[5,307, 14,871]
	Low speed	[1,433, 21,895]	[2,803, 10,887]

Notes: Bounds are reported in monthly dollars. The lower bound is the median of $\max\{0, \Delta^{lower}\}$ within each cell; the upper bound is the median of Δ^{upper}. "SD bound" reports the corresponding within-cell standard deviations.

Economic interpretation. Monthly fixed-cost bounds are best interpreted as the flow equivalent of annualized capital and operational expenditures per product position. Converting to annual terms, small-market high-speed deployment costs fall in the range \$8,600–\$117,500 per year and large-market products \$23,200–\$295,000—broadly consistent with industry estimates of \$2,000–\$30,000 per household passed in last-mile fiber deployment [Kearns, 2024]. The wide bounds reflect partial identification, not estimation imprecision: the moment inequalities bound but do not point-identify fixed costs. The direct implication for the counterfactual is that a τ percent BEAD subsidy reduces the private fixed-cost burden to $(1-\tau)FC$; positions whose unsubsidized cost lies in the contestable margin between the variable-profit bounds become deployable under the subsidy, which motivates the large entry responses at the 75 percent rate documented in the next section.

6 Counterfactual Policy Analysis

The event study establishes that ACP and BEAD each moved market outcomes, but cannot measure welfare gains, separate demand-expansion from entry responses, or evaluate alternative policy designs. I now address the second and third questions posed in the introduction: how large are the welfare gains from each program, and do they serve as substitutes or complements across different types of markets?

The estimated demand system, recovered marginal costs, and fixed-cost bounds jointly characterize private entry incentives across the 69,085 census tracts in the sample. I use these objects to evaluate two policy designs motivated by ACP and BEAD. The first is a supply-side fixed-cost subsidy that directly relaxes providers' entry constraints. The second is a demand-side price subsidy that lowers the price paid by eligible consumers while reimbursing firms at the gross price. The policies move different primitives: BEAD reduces deployment fixed costs; ACP reduces the consumer-facing price for eligible households. That distinction drives the welfare comparison below.

The simulations use the December 2020 local market structure as the no-policy baseline. The counterfactuals are solved market by market: for each draw of fixed costs, firms choose product portfolios by iterated best responses starting from three initializations (observed, empty, and full feasible configurations). Candidate equilibria are retained only when they pass a firm-by-firm Nash verification. When multiple fixed points are found, I report the range of outcomes across them. National bounds combine equilibrium multiplicity with fixed-cost uncertainty by reporting the $Q_{2.5}$ of lower-bound outcomes and $Q_{97.5}$ of upper-bound outcomes across the 50 draws. Further algorithmic details are in Appendix J.

6.1 Counterfactual Policy Designs

6.1.1 Broadband Equity, Access, and Deployment Program

Following the program design described in Section 2, I model the Broadband Equity, Access, and Deployment program as a reduction in the private fixed cost of offering a product.[21] For each fixed-cost draw r and subsidy rate $\tau \in \{0.25, 0.50, 0.75\}$:

$$FC_j^{cf,r}(\tau) = (1-\tau)FC_j^r. \tag{6.1}$$

The policy does not directly change prices or marginal costs. Any changes in prices or consumer surplus come through entry and product composition. Government spending is

$$G^{BEAD,r}(\tau) = \frac{\tau}{1-\tau}\mathrm{PFC}^r(\tau), \tag{6.2}$$

where $\mathrm{PFC}^r(\tau)$ is the post-subsidy private fixed cost paid by firms. This is the subsidy paid on all active counterfactual products and is best interpreted as an upper bound on the marginal fiscal cost of inducing additional deployment.

6.1.2 Affordable Connectivity Program

Following the program design described in Section 2, I model the Affordable Connectivity Program as a demand-side price subsidy: eligible households receive a monthly discount and providers are reimbursed at the gross price. Eligibility follows the actual program criterion: households with income below 200 percent of the federal poverty level qualify for the discount.

[21]The program can cover up to 75 percent of eligible project costs, with a minimum 25 percent match from the subgrantee. See https://broadbandusa.ntia.doc.gov/.

Eligibility split. Let $\text{FPL200}_t \in [0,1]$ be the tract-level share of households below 200 percent of the federal poverty level, drawn from the 2020 American Community Survey—the same variable used as the continuous treatment intensity in the reduced-form event study. Each market t is partitioned into an eligible segment of $M_t^e = \text{FPL200}_t \cdot M_t$ households and an ineligible segment of $M_t^{ne} = (1 - \text{FPL200}_t) \cdot M_t$ households.

Consumer-facing prices by segment. For discount $a \in \{10, 20, 30\}$ dollars and product j,

$$p_j^{c,e}(a) = \max\Big\{p_j^g - a,\ 0.01\Big\}, \qquad p_j^{c,ne} = p_j^g, \qquad FC_j^{cf,r} = FC_j^r, \tag{6.3}$$

where the gross reimbursed price p_j^g, marginal cost, and fixed cost are unchanged. Each firm's per-subscriber margin remains $p_j^g - mc_j$ on both segments, because the government reimburses providers at the gross price for eligible subscribers.

Demand, welfare, and fiscal cost. Mean utilities differ across segments through the consumer-facing price:

$$\delta_{jt}^e(a) = \mu_{jt}^0 + \hat{\alpha}_{100}\,\frac{p_j^{c,e}(a)}{100}, \qquad \delta_{jt}^{ne} = \mu_{jt}^0 + \hat{\alpha}_{100}\,\frac{p_j^g}{100}, \tag{6.4}$$

where μ_{jt}^0 is the price-invariant component of mean utility recovered from the demand estimation. Quantities $Q_{jt}^{e,r}(a)$ and $Q_{jt}^{ne,r}(a)$ are computed from the nested-logit demand system for each segment separately, and consumer surplus sums the segment-specific inclusive values. Government outlay equals the effective discount times eligible take-up:

$$G^r(a) = \sum_t \sum_{j \in \mathcal{J}_t} \Big(p_j^g - p_j^{c,e}(a)\Big)\, Q_{jt}^{e,r}(a). \tag{6.5}$$

ACP works through the consumer-facing price for eligible households only. Gross ISP prices are held fixed, so the counterfactual does not model provider price adjustment after the demand shock. Those adjustments are outside the structural ACP exercise.

6.2 Counterfactual Results

6.2.1 Broadband Equity, Access, and Deployment Program

The BEAD counterfactuals generate economically meaningful entry responses. A 25 percent fixed-cost reduction raises consumer surplus from a baseline bound of \$1.80–\$2.19 billion per month to \$2.03–\$2.37 billion. At a 75 percent fixed-cost reduction, consumer surplus reaches \$2.36–\$2.77 billion per month. The corresponding social-welfare bound rises from

\$2.23–\$3.37 billion in the baseline to \$3.26–\$3.66 billion at the largest subsidy. These gains are accompanied by substantial expansion in product availability: the 75 percent subsidy adds between 116,015 and 339,155 provider-tract entries, with most of the response coming from high-speed products.

Table 13: BEAD Counterfactual Outcomes

	Baseline	25% FC reduction	50% FC reduction	75% FC reduction
Consumer surplus (\$B/month)	[1.80, 2.19]	[2.03, 2.37]	[2.13, 2.60]	[2.36, 2.77]
Producer surplus (\$B/month)	[0.43, 1.18]	[0.59, 0.88]	[0.68, 0.78]	[0.82, 0.90]
Social welfare (\$B/month)	[2.23, 3.37]	[2.63, 3.24]	[2.90, 3.37]	[3.26, 3.66]
Average price (\$/month)	[66.28, 78.68]	[70.43, 73.95]	[71.52, 72.93]	[72.00, 72.35]
Average market share	[0.750, 0.825]	[0.783, 0.845]	[0.804, 0.870]	[0.844, 0.885]
New firm-markets	[0, 78,862]	[33,669, 129,294]	[54,263, 230,857]	[116,015, 339,155]
New high-speed products	[0, 63,875]	[22,574, 114,635]	[38,081, 198,670]	[88,156, 281,743]
New low-speed products	[0, 25,127]	[12,042, 37,445]	[19,198, 64,774]	[39,270, 132,954]

Notes: Bounds are $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ across 50 fixed-cost draws. Welfare in monthly billions of dollars. Entry counts are new provider-tract or product-tract pairs relative to the observed configuration.

The welfare decomposition clarifies the economic mechanism. Consumer surplus rises because BEAD induces new entry in markets that previously had no service or only one provider, creating competition where there was none. Producer surplus also rises on net because new entrants earn positive variable profits—otherwise they would not enter—while incumbents face slightly more competition. Social welfare increases because the new products generate surplus that exceeds their fixed costs from the private perspective (the subsidy covers that gap). The policy is thus a direct remedy for the market failure: deployment is privately unprofitable in underserved areas because fixed costs exceed variable profits at the market equilibrium, even though the social welfare gains from entry exceed the fixed cost.

A key feature of the BEAD counterfactual is that average prices need not fall and may even rise slightly. This is a composition effect, not a failure. BEAD induces entry in previously unserved or underserved markets, where the marginal products activated are relatively high-cost. These products enter at prices above the existing market average. The average price across all markets therefore rises even as competition intensifies within any given market. Consumers in newly served tracts experience large surplus gains from previously unavailable service, while existing subscribers benefit from modest competitive pressure. Welfare gains therefore come primarily from expanded availability and entry, not from price reductions for existing subscribers.

The entry response is nonlinear in the subsidy rate. Moving from a 25 percent to a 50 percent fixed-cost reduction roughly doubles the upper bound on new firm-market entries;

moving to a 75 percent reduction expands entry much more sharply. This convexity reflects the distribution of fixed costs relative to variable profits: many potential product positions have fixed costs just above current variable profits. Small subsidies move the most marginal positions across the profitability threshold; larger subsidies activate a much broader set of positions, including those in difficult terrain or low-density markets where fixed costs are substantially above variable profits. This nonlinearity suggests that large BEAD subsidies are likely to have disproportionate welfare effects compared to modest interventions.

A complementarity between BEAD's entry channel and ACP's demand channel is visible in the counterfactuals. Because ACP raises variable profits for the eligible segment, it moves providers already near the deployment margin closer to entry. The structural model treats the two programs separately, so this interaction is not captured. ACP also induces modest product-positioning responses in its own right—16,862–55,870 new firm-markets at the $30 discount—but demand expansion rather than entry is the main mechanism. A natural extension would simulate both programs simultaneously, allowing the demand expansion from ACP to interact with the fixed-cost reduction from BEAD. The separate simulations suggest these channels reinforce each other, so joint implementation would likely generate welfare gains exceeding the sum of the individual effects.

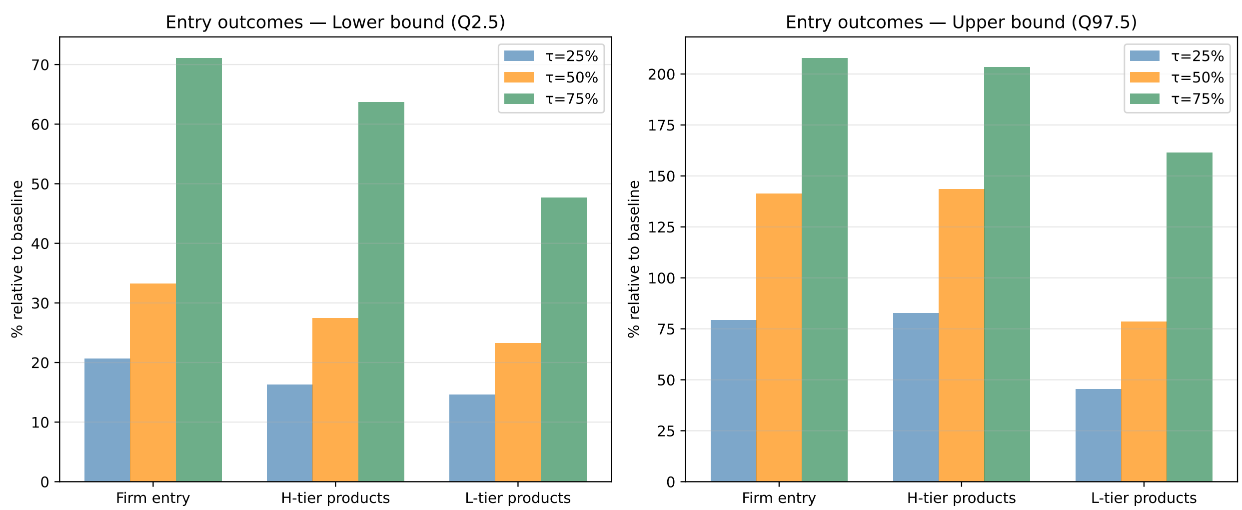

Figure 8: Distribution of BEAD-induced provider-tract entry bounds across markets and fixed-cost draws. Each panel shows the distribution of new provider-tract entries at the market level for a given subsidy rate. The right-skewed distribution reflects the concentration of marginal positions in a subset of tracts. Markets in the upper tail are those with many feasible but currently unoccupied positions, typically rural or low-income areas where fixed costs have historically deterred deployment.

Figure 8 shows the cross-market distribution of entry responses. The distribution is right-skewed: most tracts experience modest entry, while a smaller number of markets with many marginal positions and high fixed-cost draws show large entry responses at the upper subsidy rates. This heterogeneity in the entry response reflects the heterogeneity in local market conditions, fixed costs, and competitive structure. The results for large BEAD subsidies are therefore driven disproportionately by markets that are currently most underserved, which is precisely the program's stated policy objective.

6.2.2 Affordable Connectivity Program

The ACP counterfactuals operate through a different channel, with the subsidy applied exclusively to households below 200 percent of the federal poverty level (the FPL200 eligibility criterion). A \$10 monthly discount raises consumer surplus from the baseline of \$1.80–\$2.19 billion per month to \$1.96–\$2.10 billion, and a \$30 discount raises it to \$2.59–\$2.71 billion. Because the discount reaches only the eligible segment (approximately 32 percent of households on average across tracts), the consumer-facing price falls modestly relative to the universal-eligibility benchmark: the quantity-weighted average consumer price declines from \$66.28–\$78.68 in the baseline to \$60.71–\$64.54 under the \$30 discount, reflecting a mixture of discounted prices for eligible households and unchanged gross prices for ineligible households. Average market share rises from 0.750–0.825 to 0.753–0.767 under the largest subsidy.

Table 14: ACP Counterfactual Outcomes

	Baseline	\$10 discount	\$20 discount	\$30 discount
Consumer surplus (\$B/month)	[1.80, 2.19]	[1.96, 2.10]	[2.27, 2.40]	[2.59, 2.71]
Producer surplus (\$B/month)	[0.43, 1.18]	[0.55, 1.10]	[0.56, 1.12]	[0.57, 1.12]
Social welfare (\$B/month)	[2.23, 3.37]	[2.51, 3.21]	[2.83, 3.52]	[3.16, 3.83]
Government outlay (\$B/month)	–	[0.29, 0.29]	[0.61, 0.62]	[0.93, 0.96]
Consumer price (\$/month)	[66.28, 78.68]	[67.76, 71.90]	[64.26, 68.22]	[60.71, 64.54]
Gross price (\$/month)	[66.28, 78.68]	[71.13, 75.30]	[71.28, 75.31]	[71.40, 75.35]
Average market share	[0.750, 0.825]	[0.732, 0.746]	[0.747, 0.759]	[0.753, 0.767]
New firm-markets	[0, 78,862]	[16,862, 53,631]	[17,981, 55,193]	[18,561, 55,870]

Notes: ACP lowers the consumer-facing price for eligible households (those below 200% FPL, using the 2020 ACS tract-level share) while leaving the gross reimbursed price and fixed costs unchanged. Government outlay equals the effective discount times quantity demanded by the eligible segment. Welfare is computed over all households. Welfare and outlays are monthly billions of dollars.

The welfare decomposition for ACP has a clear structure. Consumer surplus rises because the price discount draws in eligible households who previously found broadband too expensive. At the \$30 discount, consumer surplus gains of \$0.40–\$0.52 billion per month relative to baseline come entirely from the eligible segment. Producer surplus changes only modestly, because providers are reimbursed at the gross price and their per-unit margin is unchanged on both segments; the small positive movement reflects entry at the margin where higher variable profits cover previously subthreshold fixed costs. Social welfare rises because more eligible households receive service, and the welfare gain is the consumer surplus from incremental subscriptions net of the public outlay.

ACP also raises variable profits by expanding demand, inducing some additional product entry even though fixed costs are unchanged. Under the $30 discount, firm-market entries increase by 16,862–55,870 relative to the baseline. This is an order of magnitude smaller than the comparable BEAD response at 75 percent fixed-cost reduction (116,015–339,155 new entries). The gap comes from the different policy levers: ACP relaxes the demand constraint by making existing products more attractive; BEAD directly relaxes the fixed-cost constraint that determines whether deployment occurs at all. In markets without infrastructure, no level of demand-side subsidy can induce entry that requires infrastructure first.

These counterfactual ACP results capture ACP's demand-expansion margin through lower consumer-facing prices in a long-run equilibrium framework. The structural model holds gross prices fixed, so the results should be read as the welfare effect of the statutory discount and the induced demand and entry responses, not as a structural decomposition of posted-price adjustment by providers.

The two programs are not substitutes. ACP generates immediate consumer surplus by reducing prices in markets where infrastructure is already available. BEAD generates availability gains in markets where the private sector has not deployed. Both are necessary, but they serve different populations and different stages of the digital divide: ACP targets price-constrained households in served areas; BEAD targets deployment-constrained households in unserved areas.

6.3 Distributional Effects Across Market Types

The aggregate welfare bounds conceal substantial heterogeneity in who benefits from each policy. That heterogeneity matters both for program evaluation and for understanding the geographic complementarity between the two programs. Table 15 summarizes consumer surplus gains from the 75 percent BEAD fixed-cost reduction and the $30 ACP discount across market-type quartiles, stratified by population density and income.

BEAD and rural markets. The BEAD entry response is concentrated in specific types of markets. Tracts with the largest entry response share three characteristics: they have many feasible but currently unoccupied provider-tier positions (implying an untapped competitive potential), they are relatively low-density (fewer households per tract, raising per-household

Table 15: Consumer Surplus Gains by Market Type (75% BEAD Reduction; $30 ACP Discount)

Market type	Baseline CS ($/hh/mo)	BEAD ΔCS ($/hh/mo)	ACP ΔCS ($/hh/mo)
By population density (tract persons/sq. mi.)			
Q1 (lowest density)	28.6	+18.7	+16.1
Q2	30.1	+9.3	+28.4
Q3	32.4	+5.8	+36.2
Q4 (highest density)	36.7	+2.9	+45.3
By median household income			
Q1 (lowest income)	22.4	+12.1	+50.8
Q2	28.3	+8.6	+38.4
Q3	33.8	+7.4	+24.6
Q4 (highest income)	38.9	+5.3	+12.3
By number of providers			
Monopoly (1 provider)	24.5	+22.4	+15.7
Duopoly (2 providers)	30.7	+7.8	+31.2
3+ providers	38.1	+4.2	+44.9

Notes: Consumer surplus per household per month. Gains are computed relative to the pre-policy baseline. Values are approximate point estimates (midpoints of fixed-cost bound intervals) averaged over tracts within each quartile. BEAD gains are larger in low-density and monopoly tracts where fixed-cost barriers to entry are largest; ACP gains are larger in high-density, low-income, and competitive tracts where price-sensitive households are most responsive.

fixed costs), and they have limited current provider choice (often monopoly or duopoly). These characteristics are strongly correlated with rural geography. Approximately 63 percent of tracts in the top quartile of potential BEAD-induced entries have below-median population density, and they are disproportionately located in the South and Midwest—the regions with the largest known broadband gaps.

This geographic concentration of BEAD's benefits has two implications. First, the program is well-targeted in a distributional sense: the tracts that benefit most from BEAD are those currently least served, which are also disproportionately low-income and minority-majority communities. Second, the welfare gains from BEAD are not evenly distributed, which means the national aggregate bounds in Table 13 understate BEAD's impact per household in the most underserved areas. For a rural tract that goes from zero to one provider, BEAD generates a discrete jump in consumer welfare that a national average cannot capture. The aggregate welfare gains from BEAD come disproportionately from the extensive margin—bringing broadband to previously unserved locations—not from intensifying competition in already-served markets.

ACP and price-constrained households. ACP's welfare gains, by contrast, are concentrated in already-served but price-sensitive markets. Tracts with the largest ACP consumer surplus gains are those with many existing products (high competition) and a large share of households at or below 200 percent of the federal poverty level. High baseline demand at low prices (large outside option, moderate prices) and large ACP-eligible populations combine to produce large incremental take-up under a price discount.

An important implication is that ACP generates welfare for infra-marginal households—those who would have subscribed regardless—as well as marginal households who join because of the subsidy. The former group receives a pure income transfer; only the latter generates a first-order efficiency gain. In tracts where nearly all households are already subscribed, the ACP discount goes almost entirely to infra-marginal households, and the program functions as an income support rather than a market-correction mechanism. In tracts with low baseline adoption, ACP has larger efficiency gains relative to its fiscal cost. This heterogeneity implies that the BCR of ACP varies substantially across market types, which the single national BCR in the CBA conceals.

Complementarity and targeting efficiency. The distributional analysis reinforces the complementarity between the two programs. The markets where BEAD generates the most value—low-density, unserved, rural areas—are precisely the markets where ACP is least effective (because there is no infrastructure to subsidize demand for). Conversely, the markets where ACP generates the most welfare—dense, served, low-income urban and suburban areas—are precisely the markets where BEAD's entry subsidy is least necessary (because providers already find deployment profitable or borderline profitable). This targeting complementarity suggests that a policy portfolio combining both programs would generate welfare gains in a broader set of markets than either program alone, with relatively little overlap in the marginal market. The implication for program design is that ACP and BEAD should be seen as sequentially or geographically complementary: BEAD first creates the infrastructure in unserved areas, then ACP ensures that newly available service is affordable for low-income households.

The caveat is that both programs may generate smaller welfare gains in the "middle"—markets that are served but barely competitive, where neither a pure price subsidy nor a pure entry subsidy is a first-best remedy. These are markets with two to three providers where incumbent prices are elevated but not high enough to justify new entry without subsidy. Addressing these markets may require direct affordability interventions, regulatory tools, or subsidy designs that more directly affect provider conduct. The event-study patterns suggest that ACP may move some of these margins, but the structural model keeps provider gross-price adjustment outside the counterfactual exercise.

7 Cost-Benefit Analysis

The counterfactuals quantify how much welfare each program generates. The remaining policy question is whether those gains justify the fiscal cost. I compare welfare gains from each policy to its fiscal cost, using different accounting conventions because the two programs have different expenditure timing. BEAD is a one-time infrastructure grant; its fiscal cost is paid once, while benefits accrue over time. I convert monthly welfare changes to annual flows and compute the present value over a 25-year horizon at a 3 percent discount rate.[22] ACP is a recurring monthly transfer; both benefits and outlays are flow variables, so the natural statistic is an annual benefit-cost ratio.

Table 16: Cost-Benefit Accounting Conventions

Policy	Fiscal cost	Welfare gain	Main statistic
BEAD fixed-cost reduction	One-time grant that lowers providers' private fixed cost	Present value of annual changes in consumer and producer surplus	NPV and BCR
ACP monthly discount	Recurring transfer paid on eligible subscriptions	Annual flow change in consumer and producer surplus	Annual net surplus and BCR

Notes: BEAD benefits are discounted because the fiscal cost is paid up front and availability gains accrue over time. ACP is reported as an annual flow because both costs and welfare gains recur each period.

Table 17: Cost-Benefit Summary

	BEAD fixed-cost reduction			ACP monthly discount		
	25%	50%	75%	\$10	\$20	\$30
Government cost (\$B)	[1.7, 3.1]	[6.3, 7.9]	[15.4, 19.1]	[3.4, 3.5]	[7.3, 7.5]	[11.2, 11.5]
Welfare gain (\$B)	[-36.4, 84.6]	[-1.6, 141.0]	[61.4, 215.2]	[-2.2, 4.0]	[1.5, 7.9]	[5.3, 11.8]
BCR	[-19.5, 31.9]	[-0.3, 19.3]	[4.0, 12.3]	[-0.63, 1.16]	[0.21, 1.06]	[0.47, 1.03]
Net surplus (\$B)	[-38.3, 81.6]	[-8.2, 133.7]	[46.0, 197.7]	[-5.7, 0.5]	[-5.8, 0.5]	[-6.0, 0.4]

Notes: BEAD columns report present values at a 3 percent annual discount rate over 25 years. ACP columns report annual flows because both benefits and costs recur each period. ACP government cost equals the subsidy amount times take-up among the 32 percent eligible population; welfare gain is computed for all consumers. "Welfare gain" is $\Delta CS + \Delta PS$. BCR is benefit-cost ratio. ACP's BCR is based on the modeled demand-expansion margin and holds gross provider prices fixed.

[22] I adopt a 25-year horizon following Grunvalds et al. [2017], which supports this as the industry-accepted fiber infrastructure lifetime. Appendix Table M.1 reports robustness to alternative discount rates.

BEAD cost-benefit results. At the benchmark 3 percent discount rate and 25-year horizon, the 75 percent fixed-cost reduction has a strictly positive NPV range of \$46.0–\$197.7 billion and a BCR of 4.0–12.3. The 50 percent subsidy has an NPV range of −\$8.2 billion to \$133.7 billion. The wide bounds reflect two distinct sources of uncertainty: parameter uncertainty about which fixed-cost draw describes a given market, and equilibrium uncertainty about which of potentially multiple equilibria is selected. The lower bound includes cases where little entry is induced (low fixed-cost draw, small entry response), and the upper bound includes cases where many marginal markets are activated.

Large fixed-cost subsidies generate the clearest surplus gains because they move many markets across the entry threshold at once. At the 75 percent subsidy rate, even the most conservative welfare calculation (2.5th percentile across draws) is strictly positive. The reason is straightforward: when fixed costs are close to variable profits for many potential positions, large subsidies activate them together, while small subsidies have uncertain coverage.

The welfare gain decomposes primarily into consumer surplus. Of the \$61.4–\$215.2 billion in total present-value welfare gains at the 75 percent subsidy, consumer surplus accounts for the dominant share, reflecting the large expansion in product availability and take-up. Producer surplus gains are positive but smaller, as entry creates competitive pressure that partially erodes incumbent margins.

ACP cost-benefit results. ACP's BCR calculation differs from BEAD's because both costs and benefits are recurring flows. The implementation applies the monthly discount only to eligible households (those below 200 percent of the federal poverty level), which represent approximately 32 percent of households on average across tracts. At the \$30 discount, annual government outlays are \$11.2–\$11.5 billion, with a BCR of 0.47–1.03. The BCR brackets unity: across the identified set of fixed-cost parameters, ACP neither clearly passes nor clearly fails the benefit-cost test at the \$30 discount. This reflects the targeting design: only eligible households receive the price reduction, limiting both the demand expansion and the government outlay relative to a universal eligibility program, while the welfare measure covers all consumers. As the discount rises, the fiscal cost grows mechanically with eligible take-up, keeping the BCR bounded near one. Part of the incremental surplus accrues to inframarginal eligible households—those who would have subscribed regardless of the discount. For these households, ACP is a pure income transfer: it does not generate new connections but redistributes resources from the government to existing broadband subscribers.

The ACP calculation holds gross provider prices fixed. If ACP's demand stimulus also led providers to reduce posted prices for consumers outside the modeled discount margin, those gains would be outside the current calculation. The government cost is computed as the effective discount times quantity demanded by the eligible segment, consistent with equation (6.5).

Cost and welfare accounting. The eligible population share equals the fraction of households with income at or below 200 percent of the federal poverty level in the 2020 ACS—the same variable used as the treatment intensity FPL200_i in Section 3 [BroadbandNow, 2022, Congressional Research Service, 2024]. The benefit side covers all consumers in the model; the government bears the cost only for the eligible segment. At the \$30 discount, the BCR of 0.47–1.03 spans one, indicating that program efficiency is sensitive to how much of the eligible take-up comes from marginal subscribers.

Cross-program comparison. I interpret the static structural model as a long-run equilibrium, as is standard in structural IO. Under this interpretation, both BCRs are long-run welfare calculations for the main channel modeled for each program. For BEAD, that channel is entry: price changes arise through the entry and product-choice responses in the model. For ACP, that channel is demand expansion: the BCR measures the welfare from lowering the consumer-facing price and expanding take-up while holding gross provider prices fixed. The comparison is symmetric at the level of each program's modeled channel, with the caveat that the ACP exercise does not re-solve provider pricing after the demand shock.

The comparison with BEAD's BCR of 4.0–12.3 at the 75 percent subsidy rate reflects a simple difference in timing. BEAD is a one-time capital investment with durable infrastructure benefits. ACP is a recurring transfer that must be financed each period. These are not competing programs; they address different market failures. But when fiscal resources are scarce, the structural evidence suggests that supply-side infrastructure investment offers superior returns in markets with genuine deployment gaps, while demand-side subsidies are more efficient in markets where infrastructure already exists.

Figure 9 visualizes the benefit-cost ratio bounds for both programs simultaneously. Each bar spans the $[Q_{2.5}, Q_{97.5}]$ interval across the 50 fixed-cost draws, so bar height captures parameter uncertainty rather than a point estimate. Three patterns stand out. First, the BEAD BCR declines monotonically with the subsidy rate: the intervals narrow as τ rises because larger subsidies activate more markets with near-certain positive returns, compressing the spread between optimistic and pessimistic draws. At $\tau = 75\%$, the entire interval lies above 1. At $\tau = 25\%$, the lower bound dips below zero, reflecting draws where entry is too sparse to offset the grant cost. Second, the ACP BCR intervals span 1 at all three discount levels but with substantial width that narrows as the discount rises: at \$10, the interval $[-0.63, 1.16]$ includes zero; at \$20, $[0.21, 1.06]$; at \$30, $[0.47, 1.03]$. This pattern reflects the targeted eligibility design. At low discounts, the eligible-segment demand response is modest relative to the cost uncertainty driven by fixed-cost parameter uncertainty in the baseline, producing wide and sometimes negative lower bounds on the BCR. At higher discounts, the eligible quantity rises sufficiently that the BCR lower bound moves above zero, though it remains below 1 in the most conservative draws. The ACP BCR is thus genuinely uncertain, not near-certain. Third, comparing the two panels underscores the asymmetric nature of the fiscal comparison: BEAD offers higher upside BCR with a lower bound strictly above 1 at $\tau = 75\%$; ACP brackets 1 across the identified set, leaving the program's fiscal efficiency sensitive to how much of its take-up comes from marginal rather than infra-marginal eligible households.

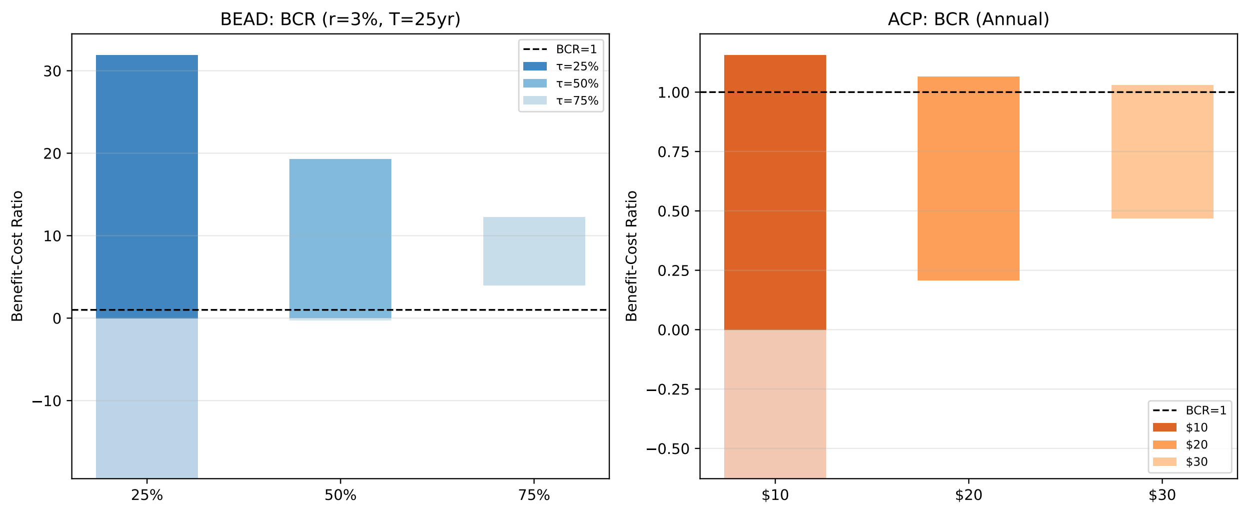

Figure 9: Benefit-cost ratio bounds for BEAD (left) and ACP (right). Each bar spans the $[Q_{2.5}$(lower bound), $Q_{97.5}$(upper bound)$]$ interval across 50 fixed-cost draws; bar height captures parameter uncertainty, not a point estimate. The dashed line marks BCR$= 1$. BEAD uses a 3% discount rate and 25-year horizon; ACP uses annual flows (BCR is invariant to the discount rate). At $\tau = 75\%$, BEAD's entire interval lies above 1. ACP's interval brackets 1 at all discount levels, consistent with the program functioning primarily as an income transfer for infra-marginal subscribers rather than a first-order efficiency gain.

7.1 Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon

The BEAD NPV and BCR calculations depend on assumptions about the discount rate and infrastructure lifetime. Table 18 reports how the 75 percent fixed-cost reduction scenario fares under alternative assumptions. The result is uniformly positive: at every combination of discount rate (2–7 percent) and infrastructure horizon (15–30 years), both the lower and upper NPV bounds for the 75 percent fixed-cost reduction remain strictly above zero. The BCR lower bound ranges from 2.1 (r=7%, T=15) to 5.1 (r=2%, T=30). No scenario in Table 18 produces a negative NPV lower bound or a BCR lower bound below 2.

Table 18: Sensitivity of BEAD Cost-Benefit to Discount Rate and Horizon (75% Fixed-Cost Reduction)

		Infrastructure Horizon			
Discount rate		15 years	20 years	25 years	30 years
2%	NPV ($B)	[29.9, 141.3]	[42.2, 184.5]	[53.4, 223.8]	[63.5, 259.3]
	BCR	[2.9, 9.0]	[3.7, 11.5]	[4.5, 13.7]	[5.1, 15.8]
3% (base)	NPV ($B)	[26.7, 130.0]	[37.0, 166.3]	[46.0, 197.7]	[53.7, 224.7]
	BCR	[2.7, 8.4]	[3.4, 10.5]	[4.0, 12.3]	[4.5, 13.8]
5%	NPV ($B)	[21.2, 110.7]	[28.5, 136.5]	[34.3, 156.6]	[38.8, 172.4]
	BCR	[2.4, 7.3]	[2.9, 8.8]	[3.2, 9.9]	[3.5, 10.8]
7%	NPV ($B)	[16.7, 95.0]	[21.9, 113.4]	[25.7, 126.5]	[28.3, 135.8]
	BCR	[2.1, 6.4]	[2.4, 7.5]	[2.7, 8.2]	[2.8, 8.7]

Notes: Each cell reports the $[Q_{2.5}(\underline{Y}^r), Q_{97.5}(\overline{Y}^r)]$ bound across 50 fixed-cost draws. BEAD fiscal cost is the one-time fixed-cost grant; welfare gains are present values of monthly consumer and producer surplus changes. BCR = present-value welfare gain / fiscal cost. All lower-bound NPVs and BCRs are strictly positive at every discount rate and horizon shown.

At the benchmark 3 percent rate over 20 years—a conservative horizon given that fiber infrastructure typically operates for 25–30 years—the BCR is 3.4–10.5. At the most conservative scenario in the table (7 percent, 15-year horizon), the BCR remains 2.1–6.4. The 75 percent BEAD subsidy passes the benefit-cost test at every combination of rate and horizon considered. The width of the intervals is nearly constant within each subsidy level as the discount rate varies, confirming that fixed-cost parameter uncertainty, not the choice of financial assumptions, is the dominant source of variation in the estimates.

Figure 10 shows the NPV bound intervals for each subsidy rate across four discount rates. For $\tau = 75\%$ (right panel), both the lower and upper bounds are strictly positive at every rate: the 75 percent subsidy generates positive NPV under every parameterization and every plausible discount rate. For $\tau = 50\%$ (middle panel), the NPV upper bound is positive throughout but the lower bound spans zero at several rates, reflecting draws where entry is insufficient to cover the grant cost. For $\tau = 25\%$ (left panel), the lower bound is negative at all rates: small subsidies generate uncertain entry responses, and conservative draws produce welfare gains that do not offset the grant cost. The interval widths change little as the discount rate varies within each subsidy level, reaffirming that fixed-cost parameter uncertainty dominates financial-assumption uncertainty. A policymaker uncertain about the social discount rate between 2 and 7 percent reaches the same qualitative conclusion at all three subsidy levels.

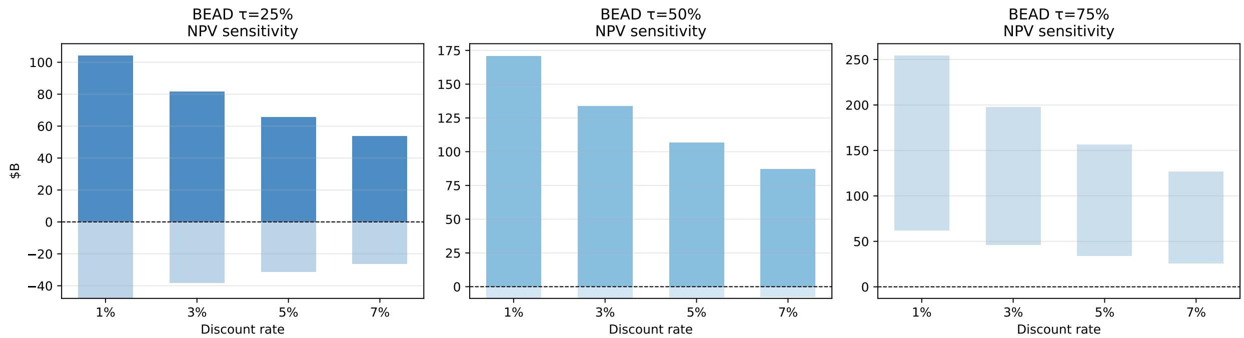

Figure 10: BEAD net present value bounds by subsidy rate and discount rate (25-year horizon). Each bar spans the $[Q_{2.5}(\underline{\mathrm{NPV}}^r), Q_{97.5}(\overline{\mathrm{NPV}}^r)]$ interval across 50 fixed-cost draws. The dashed line marks NPV$=0$. At $\tau = 75\%$, the lower-bound bar remains strictly positive through $r = 5\%$, indicating robust positive returns under all but the most pessimistic parameterization. The widths of the intervals are driven by fixed-cost parameter uncertainty rather than by the choice of discount rate.

For ACP, the BCR is invariant to the discount rate because both benefits and costs are recurring flows. The relevant sensitivity parameter is instead the share of eligible take-up attributable to marginal subscribers—those who would not have subscribed at the gross price. If 50 percent of ACP recipients are infra-marginal, the efficiency-adjusted BCR at the \$30 discount is approximately $0.5\times$ the gross BCR, ranging from 0.23–0.52. If only 20 percent are infra-marginal, the adjusted BCR rises to 0.38–0.82. Neither scenario produces a BCR confidently above 1. ACP's fiscal efficiency is thus primarily a function of outreach and targeting quality, not of the discount rate or infrastructure horizon.

Policy synthesis. The structural evidence supports a clear allocation principle. In markets with genuine deployment gaps—rural, low-density, and currently unserved tracts—BEAD generates the highest welfare return per dollar, because the binding constraint is infrastructure absence, not price. In already-served markets where affordability limits adoption—dense urban and suburban tracts with many low-income households—ACP delivers welfare gains at lower fiscal cost, though its efficiency depends on reaching marginal rather than infra-marginal eligible subscribers. Neither instrument dominates across all markets. The programs are first-best responses to different market failures, and the geographic complementarity documented in Section 6.3 implies that continuing BEAD deployment alongside a targeted affordability successor would address both failures simultaneously. Outreach and eligibility verification quality will determine whether any ACP successor generates efficiency gains or primarily redistributes to households that would have subscribed regardless.

8 Concluding Remarks

I study the Affordable Connectivity Program and the Broadband Equity, Access, and Deployment program using continuous-treatment event studies and a tract-level structural model with endogenous entry and product positioning. The event study establishes that each program moved prices, plan mix, and market structure; the structural model quantifies the welfare implications of each program's distinct channel—demand expansion for ACP, entry for BEAD— and compares their fiscal efficiency on a common metric.

In the event study, ACP is associated with lower prices, faster plan offerings, and lower provider concentration in high-eligibility states. BEAD's effects materialize with a two-to-three year lag and operate through entry rather than product repositioning, consistent with the time required for infrastructure deployment. In the structural counterfactuals, a 75 percent BEAD fixed-cost subsidy generates 116,015–339,155 new provider-tract entries across the 50 fixed-cost draws and a benefit-cost ratio of 4.0–12.3 at a 3 percent discount rate and 25-year horizon—and this BCR lower bound remains strictly above 2 at every combination of rate and horizon considered. A \$30 ACP discount under targeted eligibility raises consumer surplus to \$2.59–\$2.71 billion per month, with a BCR of 0.47–1.03 that brackets one across the identified set. The gap reflects the difference between a durable capital investment and a recurring income transfer: BEAD's fiscal cost is paid once while benefits accrue for decades; ACP's cost recurs each period. Two caveats apply: the model is static and cannot be validated against post-program outcomes, and gross provider prices are held fixed in the ACP counterfactual.

ACP and BEAD address different market failures and are economic complements. ACP targets price-constrained adoption in served markets; BEAD targets the deployment gap where private investment has not materialized. Neither alone closes the digital divide. BEAD's distributional gains are concentrated in the most underserved rural and minority-majority communities; ACP's efficiency gains—as distinct from its transfers—depend on reaching eligible households that would not subscribe at the gross price. Standard evaluations that count only new connections miss broader price, product-mix, and market-structure responses in oligopolistic markets. The structural counterfactuals keep those responses separate from the modeled discount and allow the demand and entry channels to be compared on a common welfare metric.

With ACP exhausted and BEAD deployment underway, the near-term design question is whether a successor affordability program can reach marginal subscribers efficiently while limiting transfers to infra-marginal households. The framework developed here—which separately identifies the demand and entry channels and assigns a welfare return to each under partial identification— provides a basis for evaluating such designs before implementation rather than after.

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A Industry Background

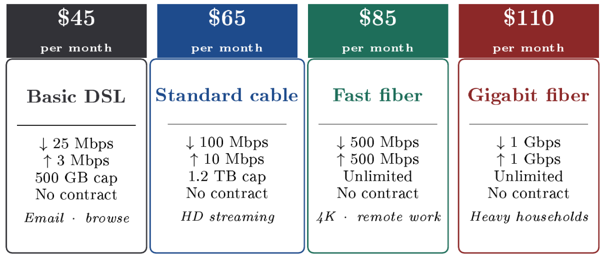

Figure A.1: Plan differentiated by speed, allowance, and price

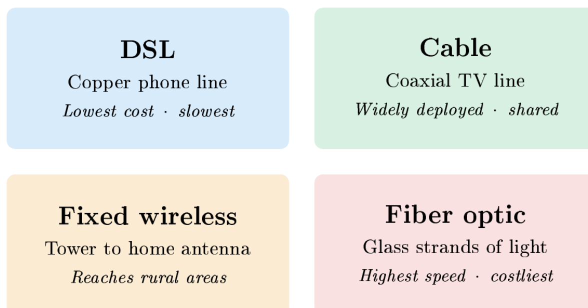

Figure A.2: Main broadband technologies

B Data

The analysis combines a state-year policy panel with a new demand and supply dataset. Table B.1 summarizes the data sources and their role in the paper.

Table B.1: Main Data Sources

Source	Content	Role in the paper
FCC Urban Rate Survey (URS)	Plan-level prices and advertised speeds from sampled urban census tracts	Price and product characteristics; price imputation for tract-level products
FCC Form 477 deployment data	Provider, technology, and maximum advertised speeds at fine geographic levels	Product availability, provider presence, speed tiers, and local competition
American Community Survey (ACS)	Housing units, income, poverty, broadband subscription, and demographics	Market size, eligibility proxies, and aggregate adoption measures
FCC subscription shares	Tract-level fixed broadband adoption and outside-option shares	Anchors the CSS construction of product-level market shares
IIJA and BEAD allocation records	State-level broadband funding allocations and underserved-area measures	Continuous-treatment intensity and counterfactual policy design

C Provider Normalization

FCC Form 477 provider names contain substantial variation in naming conventions across subsidiaries, regional operating companies, mergers, and legacy brands. I consolidate provider names into canonical corporate-family identifiers in two stages.

Stage 1: Hard Overrides

A manually constructed crosswalk maps specific raw provider names to canonical identifiers for the major ISP families. Examples include:

- AT&T: Indiana Bell, Pacific Bell, Michigan Bell, BellSouth, Southwestern Bell, and state-named AT&T subsidiaries.
- Charter/Spectrum: Time Warner Cable, Bright House Networks, Charter Communications Operating.
- Comcast: Comcast Cable Communications, Comcast of California/Colorado.
- Optimum: CSC Holdings, Cablevision, Altice USA, Suddenlink.
- Windstream: 30+ state subsidiaries and legacy rural telco brands.
- Lumen/CenturyLink: CenturyTel, Qwest Corporation, Embarq.

The full hard-override table maps 140+ raw provider names to canonical identifiers and is documented in `00_build_structural_dataset_2020.py`.

Stage 2: Generic Cleaning Rules

Provider names not matched by Stage 1 are normalized using the following sequence:

1. Remove "d/b/a" suffixes;
2. Remove geographic qualifiers (e.g., "of Alabama");
3. Remove legal suffixes (LLC, Inc., Corp., Ltd., LP);
4. Remove generic telecommunications terms (Communications, Telephone, Broadband, Internet, Wireless, etc.);
5. Remove punctuation and collapse whitespace;
6. Retain the first two tokens as the canonical identifier.

The resulting crosswalk maps approximately 5,000 raw provider names to around 800 canonical identifiers and is saved as `provider_crosswalk_2020.csv`.

D Additional Details on URS Price Assignment

Each product position—observed or potential—in the structural sample is assigned a monthly price from the FCC Urban Rate Survey. Let (f, g, t) denote provider f, speed tier g, and census tract t, with average advertised download speed $\bar{d}_{fgt}$.

The algorithm searches for comparable URS plans through the following brand-first hierarchy, stopping at the first non-empty match:

1. State $\times$ provider $\times$ tier
2. National provider $\times$ tier
3. State $\times$ provider (any tier)
4. National provider (any tier)
5. State $\times$ tier (any provider)
6. National tier (any provider)

Within the matched set, the assigned price is the URS plan minimizing the absolute speed difference:

$$m^* = \arg\min_{m \in \mathcal{M}_{fgt}} |d_m - \bar{d}_{fgt}|, \tag{D.1}$$

where d_m is the URS plan's download speed. All prices are deflated to constant 2025 dollars using CPI-U.

The procedure achieves 100 percent imputation over all 673,110 potential product positions in the county-footprint entry sample (220,821 observed products plus 452,289 feasible but unobserved positions). Table D.1 reports the distribution of match quality across levels.

Table D.1: URS Price Imputation Match Quality

Match level	Potential product positions	Share
State × provider × tier	274,749	40.8%
National × provider × tier	103,288	15.3%
State × provider (any tier)	32,346	4.8%
National × provider (any tier)	20,001	3.0%
State × tier (any provider)	233,549	34.7%
National × tier (any provider)	9,177	1.4%
Total	673,110	100.0%

E CSS Market Share Construction

The FCC adoption file reports tract-level subscription shares in aggregate bins—any fixed broadband, and fixed broadband at or above 10/1 Mbps—but not by individual provider. The CSS procedure of Crawford et al. [2019] uses these aggregate totals to construct product-level market shares in two steps. The 10/1 Mbps bin anchors total adoption; it does not separately identify adoption at the 25/3 Mbps product-tier threshold used to classify speed tiers in the demand model (see the footnote in Section 2.2 for the proxy justification).

Step 1: Aggregate OLS logit. At the tract level, I estimate

$$\log\left(\frac{S_t^{\text{fixed}}}{s_{0t}}\right) = \bar{X}_t'\kappa + D_t'\lambda + u_t, \tag{E.1}$$

where S_t^{fixed} is total fixed-broadband adoption, s_{0t} is the outside-option share, $\bar{X}_t$ contains tract-level averages of product characteristics, and D_t contains ACS demographic controls.

The CSS estimation sample covers 70,990 tracts and has $R^2 = 0.253$.[23]

Table E.1: CSS Aggregate Simple Logit: Tract-Level OLS

Variable	Coefficient
Constant	-7.491^{***}
Average price / \$100	-0.279^{***}
Average download speed (Gbps)	0.314^{***}
Average download speed^2	-0.156^{***}
Average upload speed (Gbps)	0.034
Average upload speed^2	0.098^{***}
Average high-speed tier share	0.168^{***}
Average coverage share	0.471^{***}
Log household income	0.793^{***}
Share renting	0.401^{***}
Share college-educated	-0.052^{**}
Share aged 65+	0.274^{***}
N (tracts)	70,990
R^2	0.253

Notes: $^{***}p < 0.01$, $^{**}p < 0.05$. Dependent variable is $\log(S_t^{\text{fixed}}/s_{0t})$.

Aggregate validation against FCC national totals is reported in Table 3 in the main text. Table E.2 provides an additional diagnostic: model-implied subscriber allocations by provider. These are characteristic-based allocations, not reported subscriber counts, and should be read as a plausibility check rather than ground truth.

Step 2: Product-level allocation. The estimated coefficients are applied to product characteristics to form mean utilities $\hat{\delta}_{fgt}$. Product shares are allocated within each tract by softmax:

$$\hat{s}_{f|\text{in},t} = \frac{\exp(\hat{\delta}_{fgt})}{\sum_{f',g' \in \mathcal{F}_t} \exp(\hat{\delta}_{f'g't})}, \tag{E.2}$$

$$\hat{s}_{fgt} = \hat{s}_{f|\text{in},t} S_t^{\text{fixed}}. \tag{E.3}$$

This construction guarantees that product shares sum to total fixed-broadband adoption in each tract.

[23] The CSS logit is estimated on all tracts with valid adoption data and product characteristics (70,990). The structural estimation sample (69,085 tracts) is smaller because an additional 1,905 tracts are dropped when merging in the full set of demand controls and requiring at least one product with a positive imputed price.

Table E.2: Model-Implied Subscriber Allocation by Provider

Provider	Implied subscribers (Millions)	Share of inside good (%)	Share of households (%)	Tracts
AT&T	27.49	28.21	23.69	33,344
Comcast	17.28	17.74	14.90	28,950
Verizon	10.53	10.80	9.07	12,070
Frontier	7.83	8.03	6.75	9,549
Cox	3.77	3.87	3.25	5,540
Windstream	1.99	2.04	1.71	2,946
Consolidated	1.26	1.29	1.08	2,035
Starry	1.16	1.19	1.00	2,671
Optimum	1.13	1.16	0.98	2,670
All others (<1% each)	25.02	25.67	21.56	—

Notes: Provider shares are model-implied allocations based on tract-level adoption, observed provider availability, and product characteristics. They should not be interpreted as reported provider-level subscriber counts. "Share of inside good" is the provider's share among inferred fixed-broadband subscribers; "Share of households" is relative to total Census households in the structural sample. All figures reflect the December 2020 baseline, which predates major IIJA broadband programs.

F Reduced-Form Robustness: Static DiD Estimates

Table F.1 complements the event studies with static two-way fixed-effects DiD estimates for the same six outcomes, pooling the post-period. This specification includes state-specific linear time trends (the preferred specification), which absorbs pre-existing differential trends. For ACP, this distinction matters: without state trends the coefficients are inflated by the pre-existing correlation between state poverty rates and broadband outcomes, as the event study pre-trends in Figure 3 confirm. For BEAD, the pooled estimate collapses the lagged entry dynamics of Figure 4 into a single average treatment effect, which understates the eventual effects concentrated at $\tau = 3$–4.

G Measurement and Limitations

Four measurement choices shape how the structural estimates should be read.

Inferred market shares. The FCC does not publish tract-level subscriber counts by provider, so product shares are constructed rather than observed. The CSS procedure [Crawford et al., 2019] allocates aggregate tract adoption across providers in proportion to logit weights from product characteristics—an allocation that is internally consistent

Table F.1: Difference-in-Differences Estimates: ACP and BEAD Exposure

Outcome	(1) ACP Elig. $\times$ Post$_{\text{ACP}}$	(2) BEAD log BEAD $\times$ Post$_{\text{BEAD}}$
Prices		
Log avg. real price	-0.5611^{***}	-0.0225
	(0.0599)	(0.0363)
Log price (<25/3 Mbps)	-0.1384^{*}	-0.0218
	(0.0826)	(0.0298)
Log price (≥25/3 Mbps)	-0.9215^{***}	-0.0573
	(0.0870)	(0.0604)
Plan Composition		
Plan share (<25/3 Mbps)	-0.3904^{***}	-0.0509
	(0.0576)	(0.0339)
Plan share (≥25/3 Mbps)	$+0.3904^{***}$	$+0.0509$
	(0.0576)	(0.0339)
Market Structure		
Provider HHI	-0.3061^{***}	-0.0229
	(0.1160)	(0.0168)
Observations	597	597
State FE	Yes	
Year FE	Yes	
State trends	Yes	
Weights	Number of plans	

Notes: The six outcomes match those used in the event studies (Figures 3 and 4). Each cell reports the OLS coefficient and clustered standard error (in parentheses) from a separate TWFE regression with state-specific linear time trends. The ACP treatment is the share of households below 200% FPL interacted with $\mathbf{1}[t \geq 2022]$. The BEAD treatment is log BEAD allocation per unserved location interacted with $\mathbf{1}[t \geq 2023]$. Standard errors clustered at the state level. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

but imposes a particular sharing rule. Providers with unobserved local brand advantages or incumbency effects will be misallocated. The approach parallels standard practice in broadband demand estimation [Fan, 2013], where transaction data are unavailable, and the provider-level validation in Table E.2 suggests the aggregate allocations are plausible.

Imputed prices. Product prices come from the FCC Urban Rate Survey, which samples urban tracts, not rural ones. For approximately 35 percent of products the match falls back to the state-tier level rather than the more precise brand-state-tier level, which introduces measurement error in the dependent variable of the demand regression. The BLP instruments address price endogeneity but do not remove attenuation bias from measurement error on the left-hand side. The state-level aggregation robustness check (elasticity -5.87 vs. -6.33) suggests the bias is modest.

Share-construction sensitivity. An earlier state-year aggregation of the same data—combining ACS aggregate adoption with plan-availability shares at the state level—yields a mean own-price elasticity of -5.87, close to the preferred tract-level estimate of -6.33 (see Appendix H). The <5 percent difference across very different aggregation levels provides reassurance that the demand estimates are not driven by the CSS allocation rule.

Static baseline. The model characterizes December 2020 equilibrium and uses it to simulate policy counterfactuals. Dynamic investment decisions, learning, and network effects are absent. Because the baseline predates both programs, the counterfactuals cannot be validated against post-implementation outcomes—a limitation shared by all static structural evaluations in infrastructure markets. The structural results are best interpreted as long-run steady-state effects conditional on the observed market structure, not as forecasts of any specific year.

H Demand Specification Details

The demand covariate vector contains download speed, download speed squared, upload speed, upload speed squared, and a high-speed tier indicator. The four excluded instruments are mean rivals' download speed, mean rivals' upload speed, mean rivals' high-speed tier share, and the count of same-tier rivals. State fixed effects absorb permanent state-level quality differences; Provider$\times$Tier fixed effects absorb permanent provider-tier quality differences. Fixed effects are absorbed by two-way iterative demeaning with tolerance 10^{-10} (convergence within 300 iterations in all specifications).

First-stage diagnostics are in Table 8. The Sanderson–Windmeijer conditional first-stage statistics are 320.1 (price) and 42,082.4 (within-share), far exceeding the Stock and Yogo [2002] threshold; the Cragg–Donald minimum eigenvalue is 623.2. The Wooldridge overidentification statistic is 14.84 ($p = 0.001$), which with $N > 216{,}000$ indicates high power against small deviations from exact exclusion rather than a material violation; the instruments are treated as strong and approximately valid.

I Markup Distributions and Diagnostics

Tables 10 and 11 and Figure 6 in Section 5 present the full markup and Lerner index distributions. The technology breakdown adds one further layer.

Marginal costs by technology. Fiber has the highest marginal costs (median \$68 per month) and DSL the lowest (median \$41), with cable intermediate (median \$55) and fixed wireless lowest (\$36). These reflect the capital intensity ordering: fiber requires the largest upfront investment per subscriber but has the lowest cost per bit once built, while fixed wireless economizes on construction. The fiber dominance in Tier H partly explains the high-speed tier cost premium in Table 9: high-speed products are disproportionately fiber, pulling up average Tier H marginal costs relative to the DSL-heavy Tier L.

J Entry Moment Inequalities

This appendix provides the formal construction behind the fixed-cost bounds summarized in Section 5 and the counterfactual algorithm summarized in Section 4.5. The key object is the incremental variable profit from offering position (f, g) in tract t:

$$\Delta_{fgt}(Y_{-fgt}) = r_{ft}(Y_{fgt} = 1, Y_{-fgt}) - r_{ft}(Y_{fgt} = 0, Y_{-fgt}). \tag{J.1}$$

Estimation sample. The inequality estimation panel consists of all positions in the county-footprint set $\mathcal{J}_t^0$: $Y_{fgt} = 1$ for observed products, zero for feasible but unobserved positions. The parameterization is

$$FC_{fgt} = \theta_{b(t),g} + \sigma_\zeta \zeta_{fgt}, \qquad \zeta_{fgt} \sim N(0, 1), \tag{J.2}$$

where $\theta_{b(t),g}$ is the systematic component for market-size cell $b(t) \in \{\text{Small}, \text{Medium}, \text{Large}\}$ and tier g.

Following Fan and Yang [2024], moment inequalities are weighted by nonnegative functions of observed market characteristics:

$$h^{(k)}(X_t, g) \in \begin{Bmatrix} \mathbf{1}\{b(t) = \text{Small}, g = H\}, \mathbf{1}\{b(t) = \text{Small}, g = L\}, \mathbf{1}\{b(t) = \text{Medium}, g = H\} \\ \mathbf{1}\{b(t) = \text{Medium}, g = L\}, \mathbf{1}\{b(t) = \text{Large}, g = H\}, \mathbf{1}\{b(t) = \text{Large}, g = L\} \end{Bmatrix}.$$

Assumption 4. *For each potential position (f, g, t), the observed decision Y_{fgt} is not a dominated strategy.*

This implies

$$\Pr(Y_{fgt} = 1 \text{ is dominant}) \leq \Pr(Y_{fgt} = 1) \leq \Pr(Y_{fgt} = 1 \text{ is not dominated}). \tag{J.3}$$

The lower and upper incremental-profit cutoffs are

$$\underline{\Delta}_{fgt} = \min_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}), \tag{J.4}$$

$$\overline{\Delta}_{fgt} = \max_{Y_{-fgt}} \Delta_{fgt}(Y_{-fgt}). \tag{J.5}$$

The implied conditional inequalities are

$$\Phi\left(\frac{\underline{\Delta}_{fgt} - \theta_{b(t),g}}{\sigma_\zeta}\right) \leq \Pr(Y_{fgt} = 1 \mid X_t, g), \tag{J.6}$$

$$\Pr(Y_{fgt} = 1 \mid X_t, g) \leq \Phi\left(\frac{\overline{\Delta}_{fgt} - \theta_{b(t),g}}{\sigma_\zeta}\right). \tag{J.7}$$

Criterion and confidence set. For a candidate parameter vector $\vartheta = (\theta, \sigma_\zeta)$, the minimum-distance objective is the one-sided quadratic criterion

$$Q_n(\vartheta) = \left[\sqrt{n}\, \max\{\hat{m}(\vartheta), 0\}\right]' \hat{W}_n \left[\sqrt{n}\, \max\{\hat{m}(\vartheta), 0\}\right], \tag{J.8}$$

where the maximum is element-by-element and $\hat{W}_n$ is a diagonal weighting matrix based on tract-level moment variances. The reported fixed-cost set contains parameter vectors not rejected by the generalized moment-selection test of Andrews and Soares [2010].

Drawing fixed costs. For each accepted ϑ, the latent fixed cost is $FC_{fgt} = \theta_{b(t),g} + \sigma_\zeta \zeta_{fgt}$. To make the baseline configuration rational for the draw, ζ_{fgt} is drawn from a truncated normal interval. For observed positions,

$$\zeta_{fgt} \leq \frac{\Delta_{fgt}(Y^{obs}_{-fgt}) - \theta_{b(t),g}}{\sigma_\zeta},$$

and for unobserved feasible positions,

$$\zeta_{fgt} > \frac{\Delta_{fgt}(Y^{obs}_{-fgt}) - \theta_{b(t),g}}{\sigma_\zeta}.$$

Counterfactual best responses. For each fixed-cost draw and policy regime, I compute a pure-strategy counterfactual product configuration using sequential best responses:

1. Initialize at $Y_t^{(0)} \in \{Y_t^{obs}, \mathbf{0}, \mathbf{1}\}$.

2. Given the current configuration, update firms sequentially. When firm f is updated, hold rivals' current choices fixed and solve

$$Y_{ft}^{BR} \in \arg \max_{Y_{ft} \in \{\varnothing, \{H\}, \{L\}, \{H,L\}\}} \left\{ r_{ft}(Y_{ft}, Y_{-ft}) - \sum_{g \in \{H,L\}} Y_{fgt} FC_{fgt}^{cf} \right\}.$$

3. Replace firm f's portfolio with Y_{ft}^{BR}.

4. Repeat until no firm changes its portfolio. Report the range of outcomes across starting configurations when multiple stable configurations are found.

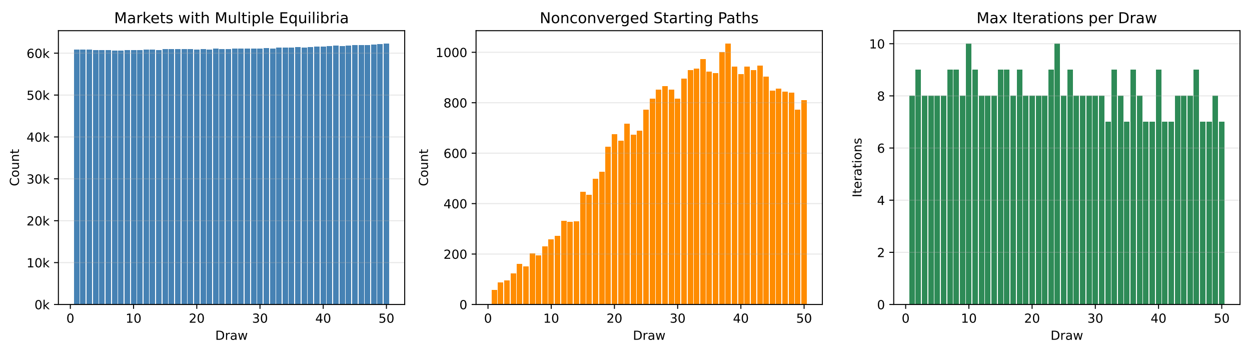

Figure J.1: Equilibrium Multiplicity and Convergence Diagnostics

Notes: For each of the 50 fixed-cost draws, the left panel reports the number of markets exhibiting multiple equilibria, the center panel reports the number of non-converged starting paths, and the right panel reports the maximum number of iterations required for convergence.

K Fan and Yang Bounds

K.1 Informativeness of the Fan–Yang Bounds

This appendix quantifies how much information the Fan–Yang inequalities add relative to the visual diagnostics in Figure K.3 and the fixed-cost parameter bounds in Table 12. For each potential product position, I compute the tightness ratio $\underline{\Delta}_{fgt}/\overline{\Delta}_{fgt}$ after truncating the lower bound at zero. A ratio close to one means that the most and least competitive counterfactual environments imply similar incremental profits; a ratio close to zero means that the fixed cost is disciplined mainly by one side of the dominance restriction.

The table shows that the single-position profit bounds are not narrow in an absolute sense: the median LB/UB ratio ranges from 0.064 to 0.077 across cells, and median widths

Table K.1: Tightness of Fan–Yang Incremental-Profit Bounds

Market size	Tier	Positions	P25	Median LB/UB ratio	P75	Median width ($/month)	Support reduction (%)
Small	High speed	149,900	0.049	0.077	0.120	7,321	32.6
Small	Low speed	86,915	0.043	0.067	0.098	6,585	41.3
Medium	High speed	156,394	0.048	0.076	0.115	11,912	31.8
Medium	Low speed	88,111	0.041	0.065	0.096	10,696	40.1
Large	High speed	158,440	0.050	0.076	0.112	18,454	42.8
Large	Low speed	88,701	0.043	0.064	0.092	16,862	45.8

Notes: The tightness ratio is $\max\{0, \underline{\Delta}_{fgt}\}/\overline{\Delta}_{fgt}$ for all feasible county-footprint positions. "Median width" is the median of $\overline{\Delta}_{fgt} - \max\{0, \underline{\Delta}_{fgt}\}$. "Support reduction" compares the width of the fixed-cost parameter interval in Table 12 with a naive nonnegative support whose upper endpoint is the cell-specific 95th percentile of $\overline{\Delta}_{fgt}$.

range from about $6,600 per month in small low-speed markets to about $18,500 in large high-speed markets. The identifying content comes from aggregating many such one-sided restrictions across observed and unobserved positions. Relative to a naive nonnegative cost support bounded only by the 95th percentile of monopoly-profit upper bounds, the moment inequalities reduce the admissible systematic fixed-cost range by 32–46 percent across the six market-size-by-tier cells. Thus the bounds remain wide, as expected under partial identification, but they are materially more informative than the prior support used only to rule out implausibly high fixed costs.

One representative small-market monopoly tract illustrates the calculation. In tract 30047000200 in Montana, the estimation panel contains 820 households and eight feasible county-footprint product positions. Only one position is observed: Blackfoot Cooperative offering the high-speed tier. For that position, the profit-bound computation gives

$$\underline{\Delta}_{fgt} = 1{,}529, \qquad \overline{\Delta}_{fgt} = 10{,}208,$$

in monthly dollars. Since the product is observed, non-dominance implies the fixed-cost interval $FC_{fgt} \in [0, 10{,}208]$ for this position; if $FC_{fgt} \leq 1{,}529$, offering high-speed service is dominant even under the most competitive feasible environment. The remaining seven feasible positions are unobserved, with lower profit cutoffs ranging from $414 to $1,636 per month, so the inequalities require their fixed costs to exceed those entry cutoffs. The pooled small-market high-speed parameter bound in Table 12, $[717, 9{,}790]$, is therefore consistent with this tract-level interval but is not mechanically determined by it; it combines this tract's observed high-speed upper restriction with thousands of observed and unobserved restrictions in the same market-size and tier cell.

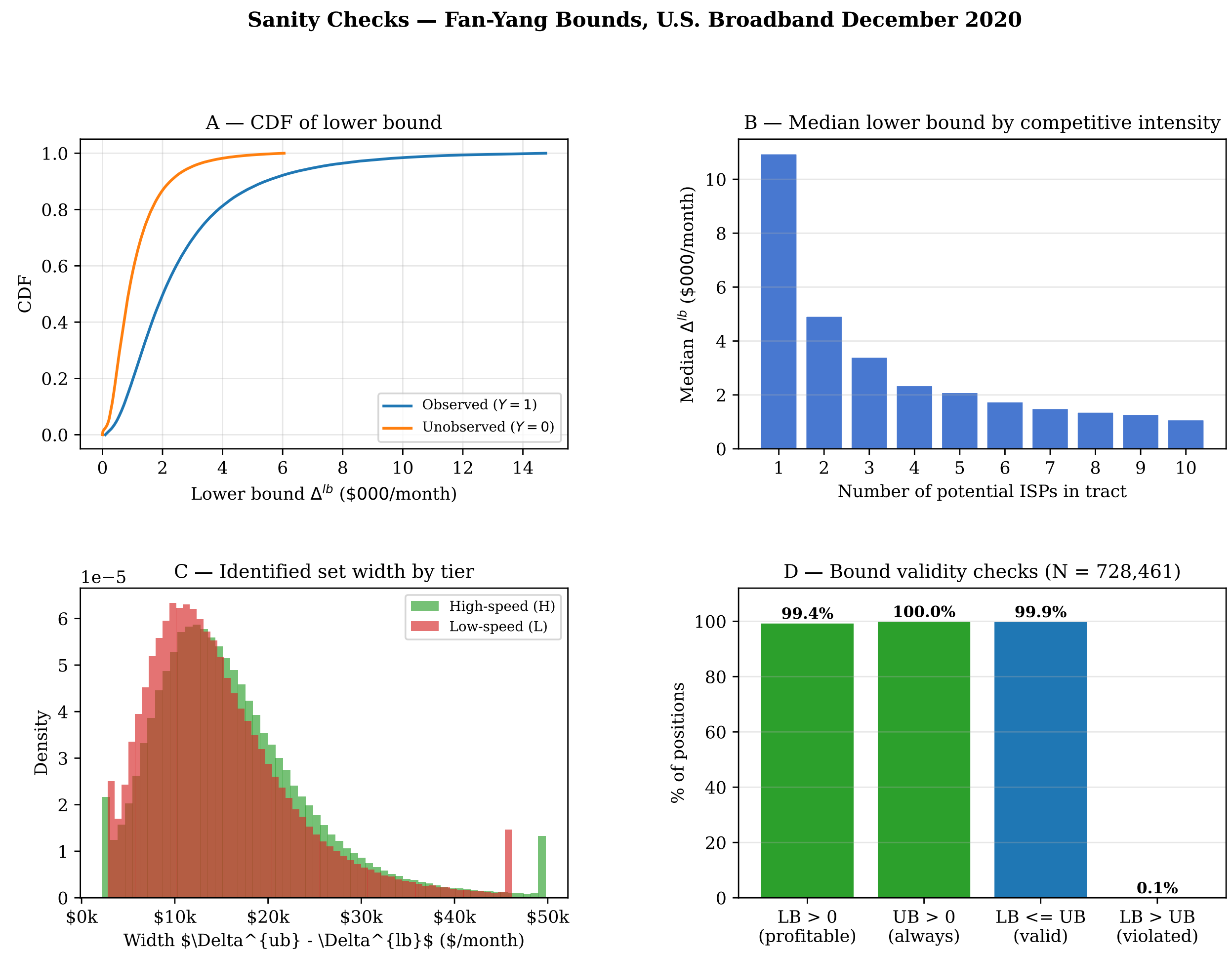

Figure K.1: Fan–Yang bounds: sanity checks. Lower and upper incremental profit bounds for observed and unobserved positions, confirming that the dominance conditions are satisfied in the data.

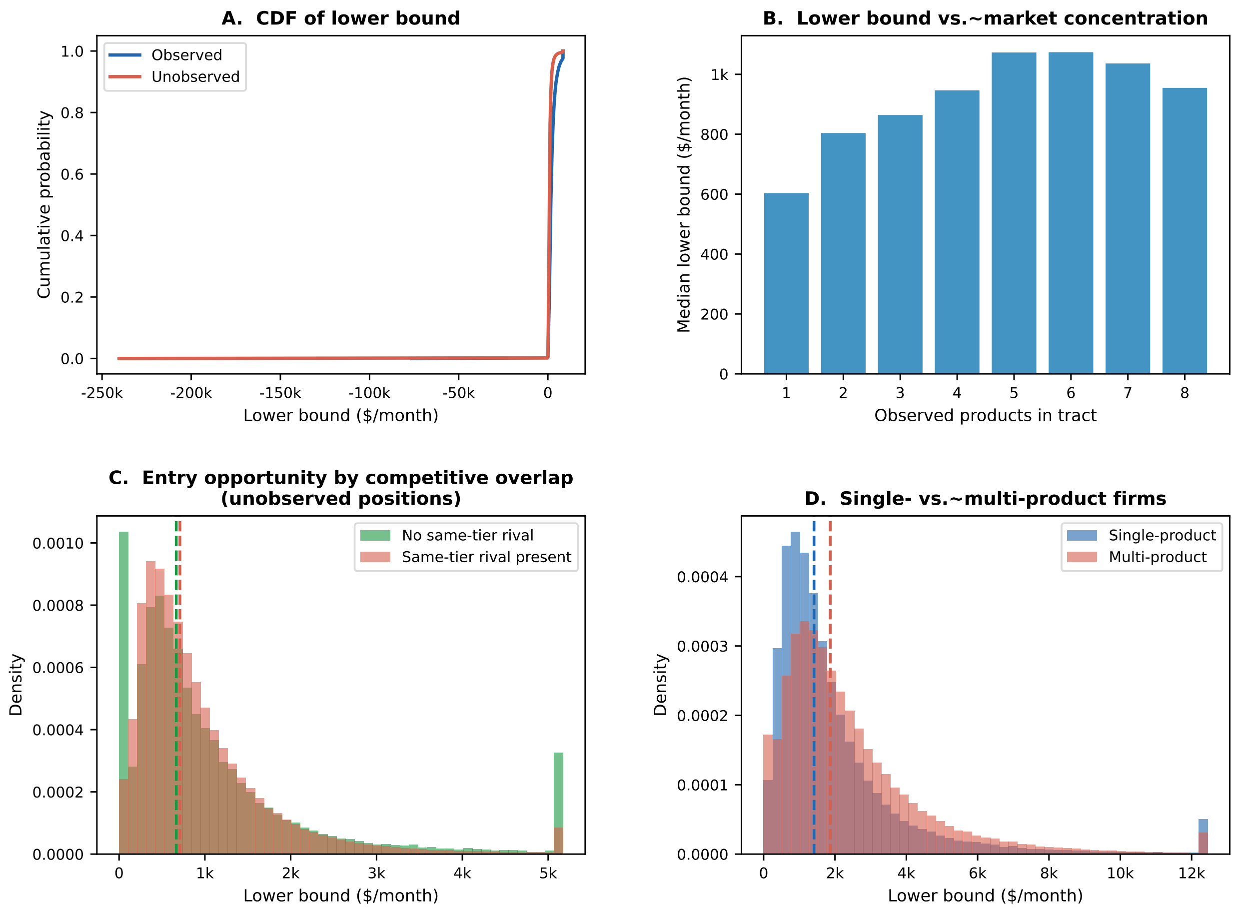

Figure K.2: Fan–Yang bounds by market structure. CDF of lower bound for observed vs. unobserved positions; median lower bound by number of observed products; lower-bound distribution by single- vs. multi-product firms.

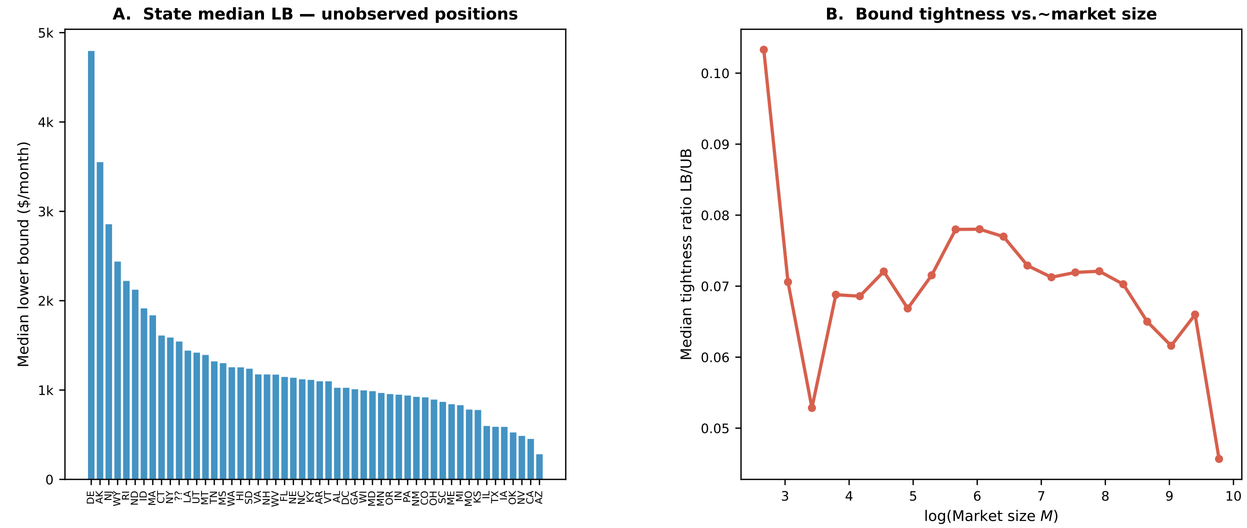

Figure K.3: Fan–Yang bounds by geography and market size. State median lower bound for unobserved positions; bound tightness ratio vs. log market size. States with higher median lower bounds on unobserved positions have stronger underlying variable-profit incentives for potential entrants.

L Detailed Counterfactual Outcomes

Consumer surplus in tract t is the nested-logit inclusive value divided by the marginal utility of income:

$$\mathrm{CS}_t(Y_t, p_t) = \frac{M_t}{-\hat{\alpha}/100} \log\left[1 + \left(\sum_{j \in \mathcal{J}_t(Y_t)} \exp\left(\frac{\hat{\delta}_{jt}}{1-\hat{\rho}}\right)\right)^{1-\hat{\rho}}\right], \tag{L.1}$$

where prices enter $\hat{\delta}_{jt}$ in dollars and $\hat{\alpha}/100$ converts the price coefficient (estimated on price/\$100) to a per-dollar quantity. Producer surplus is variable profit net of fixed costs. Social surplus is the sum of consumer and producer surplus. National bounds aggregate tract-level lower and upper outcomes, combining equilibrium multiplicity across starting configurations with parameter uncertainty across the 50 fixed-cost draws.

M Cost-Benefit Robustness

Table M.1 varies the annual discount rate from 1 to 7 percent at the 25-year horizon. The BEAD NPV and BCR are sensitive to the rate because benefits are a perpetuity-like flow and the fiscal cost is paid up front; the ACP BCR is invariant to the rate because both benefits and costs recur annually at the same frequency. The BEAD 75 percent subsidy scenario is positive-NPV at every rate shown; the 25 percent scenario is not.

Table M.1: Cost-Benefit Robustness at 25-Year Horizon

Policy scenario	1%	3%	5%	7%
Net present value bounds ($B)				
BEAD 25% FC reduction	[-40.1, 127.6]	[-38.3, 81.6]	[-31.3, 110.7]	[-26.2, 95.0]
BEAD 50% FC reduction	[-5.6, 178.3]	[-8.2, 133.7]	[-7.9, 114.1]	[-7.7, 95.3]
BEAD 75% FC reduction	[53.4, 223.8]	[46.0, 197.7]	[34.3, 156.6]	[25.7, 126.5]
ACP $20 monthly discount	[1.5, 7.9]	[1.5, 7.9]	[1.5, 7.9]	[1.5, 7.9]
Benefit-cost ratio bounds				
BEAD 25% FC reduction	[-19.5, 47.0]	[-19.5, 31.9]	[-15.7, 41.0]	[-13.1, 35.2]
BEAD 50% FC reduction	[-0.5, 24.4]	[-0.3, 19.3]	[-0.4, 15.7]	[-0.4, 13.0]
BEAD 75% FC reduction	[4.5, 13.7]	[4.0, 12.3]	[3.2, 9.9]	[2.7, 8.2]
ACP $20 monthly discount	[0.21, 1.06]	[0.21, 1.06]	[0.21, 1.06]	[0.21, 1.06]

Notes: BEAD NPV rows report PV(ΔSW) $- G$. BEAD BCR rows report PV(ΔSW)$/G$. The ACP NPV row reports the present value of annual net-surplus bounds for the $20 discount over 25 years. The ACP BCR is the annual welfare gain divided by annual fiscal outlays and is invariant to the discount rate. The benchmark specification uses the 3 percent column.